

Dittert's Conjecture in Dimensions Six through Fifteen

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Abstract

Let K_n be the simplex of nonnegative $n \times n$ matrices whose entries sum to n . If $A \in K_n$ has row sums $r_1, \dots, r_n$ and column sums $c_1, \dots, c_n$, define

$$\Phi(A) = \prod_{i=1}^n r_i + \prod_{j=1}^n c_j - \text{per}(A).$$

Dittert's conjecture asserts that Φ is uniquely maximized by the uniform matrix $U_n = n^{-1}J_n$, with maximum $2 - n!/n^n$. We prove the conjecture for every $6 \leq n \leq 15$. A Hall-deficit dilation converts a hypothetical boundary maximizer into a doubly stochastic matrix with a forced zero pattern. Stable-polynomial capacity yields support-sensitive permanent bounds and entropy refinements of the sharp rectangular-zero-block inequality. A complementary-minor entropy theorem resolves the one-unit Hall cores in dimension seven. Dimension six requires a broader core-marginal entropy estimate, an exact core-scaling stationarity identity, and a pairwise smoothing reduction to finitely many four-variable rational polynomials. Their positivity, together with the remaining scalar inequalities, is certified by exact rational Bernstein subdivision. An independent pairwise-averaging argument proves that every entrywise positive global maximizer is uniform. The complete exact-arithmetic verification program is supplied with the paper.

Keywords. Permanent; Dittert conjecture; doubly stochastic matrix; stable polynomial; polynomial capacity; Hall condition; Birkhoff polytope.

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1 Introduction

For an $n \times n$ nonnegative matrix $A = (a_{ij})$, let r_i and c_j denote its row and column sums. Dittert proposed the extremal problem of maximizing

$$\Phi(A) = \prod_i r_i + \prod_j c_j - \text{per}(A)$$

on the simplex K_n of matrices whose entries have total sum n . The conjectured unique maximizer is the uniform matrix $U_n = n^{-1}J_n$. On the doubly stochastic polytope the two product terms equal one, so Dittert's assertion reduces to the van der Waerden permanent theorem of Egorychev and Falikman [2, 3]. The conjecture was recorded by Minc [9] and later by Zhan [13].

Sinkhorn proved the case $n = 2$ [11], and Hwang proved the case $n = 3$ [6]; Hwang had earlier shown that an entrywise positive global maximizer must be uniform [5]. Cheon and Wanless developed boundary-exclusion methods and other structural results [1]. Udayan and

Somasundaram proved the case $n = 4$ [12]. Pang proved all dimensions $n \geq 17$ [10], and Kafidov proved $n = 16$ [7]. The present paper proves the ten dimensions

$$6 \leq n \leq 15.$$

The proof separates the interior and boundary of K_n . The interior classification is dimension-free: pairwise averaging is exactly flat at a positive global maximizer, whereas the uniform matrix is a strict local maximizer. Cyclic averaging and a backward strictness argument force every positive global maximizer to be uniform.

For a boundary maximizer, the row- and column-product losses share one deficit budget. Relative entropy controls subset sums, and a max-flow/min-cut criterion shows that a scalar dilation dominates a doubly stochastic matrix B . At a critical cut, B contains a rectangular zero block. Stable-polynomial capacity gives permanent lower bounds adapted to that support.

Dimensions 8, 9, 10 require a second-order scalar comparison and a separate analysis of one-sided Hall cuts. Dimensions 11, 12, 13, 14 are handled by first-order margins, apart from one deficient-column configuration in dimension 11, and dimension 15 admits a thin-thick dichotomy. Dimension seven is qualitatively different: three critical cuts with $n - a - b = 1$ survive the scalar method. A geometric-mean entropy inequality for complementary permanent minors strengthens the sharp zero-block bound on these one-unit cores. Stationarity and conditional product energy then reduce the remaining cases to three exact two-variable polynomial alternatives.

Dimension six requires one further level of structure. Critical cores with $n - a - b \geq 2$ can survive all scalar comparisons. We prove a core-marginal entropy refinement of the rectangular-zero-block bound that records both normalized marginal distributions of the Hall core. An exact core-scaling identity couples this entropy to the row and column products. A pairwise smoothing lemma reduces each side to a uniform distribution on its support and two levels of row or column sums. The six Hall block types then become 98 explicit four-variable rational polynomials. One-sided cuts are excluded by retaining the exact complementary column-product loss.

The finite positivity statements are verified by tensor-product Bernstein subdivision with exact rational coefficients. This is a computer-assisted proof component, not a floating-point experiment: every accepted box has strictly positive Bernstein coefficients, and every subdivision is exact de Casteljau averaging. The dimension-seven-through-fifteen program uses only the Python standard library; the dimension-six generator uses `sympy.Rational` for polynomial expansion and NumPy object arrays as containers, with all comparisons performed by `fractions.Fraction`. The complete source code reproduces every certificate.

Two ingredients often taken from the literature are proved here rather than used as black boxes. We reconstruct the derivative-capacity mechanism associated with Gurvits [4], deriving all permanent bounds needed below, and we give an independent proof of Hwang’s positive-maximizer conclusion. Combined with the results cited above, the theorem proved here leaves only dimension 5 outside the established range discussed in this paper.

2 Notation and the main theorem

Let S_n be the symmetric group. For $A = (a_{ij}) \in \mathbb{R}_{\geq 0}^{n \times n}$,

$$\text{per}(A) = \sum_{\sigma \in S_n} \prod_{i=1}^n a_{i, \sigma(i)}.$$

Let

$$K_n = \left\{ A \in \mathbb{R}_{\geq 0}^{n \times n} : \sum_{i,j=1}^n a_{ij} = n \right\}.$$

For $A \in K_n$, put

$$r_i = \sum_{j=1}^n a_{ij}, \quad c_j = \sum_{i=1}^n a_{ij},$$

and

$$\Phi(A) = \prod_{i=1}^n r_i + \prod_{j=1}^n c_j - \text{per}(A).$$

Let $J_n = \mathbf{1}\mathbf{1}^\top$, let

$$U_n = \frac{1}{n} J_n,$$

and define

$$\gamma_n = \text{per}(U_n) = \frac{n!}{n^n}.$$

A matrix is called *positive* if every entry is strictly positive. A matrix is *doubly stochastic* if every row sum and every column sum equals one.

Theorem 2.1 (Main theorem). *For every integer n with $6 \leq n \leq 15$ and every $A \in K_n$,*

$$\Phi(A) \leq 2 - \frac{n!}{n^n}.$$

Equality holds if and only if $A = U_n$.

The simplex K_n is compact and Φ is continuous, so global maximizers exist. We first develop two pieces of infrastructure that are valid in arbitrary dimension: permanent bounds from capacity and the classification of positive maximizers.

3 Stable-polynomial capacity and permanent bounds

3.1 Stability and capacity

Let

$$\mathbb{H} = \{z \in \mathbb{C} : \text{Im } z > 0\}.$$

A polynomial $p \in \mathbb{R}[x_1, \dots, x_m]$ is *real stable* if

$$p(z_1, \dots, z_m) \neq 0 \quad (z_1, \dots, z_m \in \mathbb{H}).$$

For a homogeneous polynomial p of degree m in m variables with nonnegative coefficients, define

$$\text{Cap}(p) = \inf_{x_1, \dots, x_m > 0} \frac{p(x_1, \dots, x_m)}{x_1 \cdots x_m}.$$

Set

$$G(1) = 1, \quad G(k) = \left(\frac{k-1}{k} \right)^{k-1} \quad (k \geq 2),$$

and

$$v_m = \frac{m!}{m^m} \quad (m \geq 1), \quad v_0 = 1.$$

The telescoping identity

$$\prod_{k=1}^m G(k) = v_m \tag{3.1}$$

will be used repeatedly. The sequence $G(k)$ is decreasing. Indeed, $G(k)^{-1} = (1 + 1/(k-1))^{k-1}$, and the function

$$x \mapsto x \log(1 + 1/x)$$

is increasing on $(0, \infty)$ because $\log(1 + y) > y/(1 + y)$ for $y > 0$.

We use two classical closure properties of real stability. They are included for completeness.

Lemma 3.1 (Closure under specialization and differentiation). *Let p be real stable.*

- (i) *Specializing one variable to a real number produces either a real-stable polynomial or the zero polynomial.*
- (ii) *Every nonzero partial derivative of p is real stable.*

Proof. For real specialization, fix $a \in \mathbb{R}$ and replace it first by $a + i\varepsilon$. The resulting polynomials are zero-free on the product of upper half-planes and converge locally uniformly as $\varepsilon \downarrow 0$. Hurwitz's theorem implies that the limit is either zero-free there or identically zero.

For differentiation, write $p(z', w) = \sum_{r=0}^d a_r(z') w^r$ with a_d not identically zero. The functions $(iR)^{-d} p(z', iR)$ are zero-free for $z' \in \mathbb{H}^{m-1}$ and converge locally uniformly to $a_d(z')$ as $R \rightarrow \infty$. Hurwitz's theorem shows that a_d is real stable, so $a_d(z') \neq 0$ for every $z' \in \mathbb{H}^{m-1}$. Thus, after fixing such a z' , the univariate polynomial $w \mapsto p(z', w)$ has exact degree d , and every one of its roots lies in the closed lower half-plane. By the Gauss–Lucas theorem, the roots of its derivative lie in the convex hull of those roots and hence also in the closed lower half-plane. Therefore the derivative does not vanish for $w \in \mathbb{H}$. Since z' was arbitrary, the partial derivative is real stable. The zero-polynomial case is excluded by hypothesis. $\square$

The next one-variable estimate is the quantitative core of the capacity method.

Lemma 3.2 (One-variable capacity estimate). *Let f be a nonzero univariate polynomial with nonnegative coefficients. Suppose all zeros of f are real and nonpositive, and let $d = \deg f \geq 1$. Then*

$$f'(0) \geq G(d) \inf_{t>0} \frac{f(t)}{t}. \tag{3.2}$$

Proof. Assume first that $f(0) > 0$. Then

$$f(t) = f(0) \prod_{\ell=1}^d (1 + \lambda_\ell t), \quad \lambda_\ell \geq 0.$$

Let

$$s = \sum_{\ell=1}^d \lambda_\ell = \frac{f'(0)}{f(0)}.$$

Since $d \geq 1$ is the exact degree, $s > 0$. By AM–GM,

$$\prod_{\ell=1}^d (1 + \lambda_\ell t) \leq \left(1 + \frac{st}{d}\right)^d.$$

If $d \geq 2$, choose $t = d/[s(d-1)]$. This gives

$$\inf_{t>0} \frac{f(t)}{t} \leq \frac{f'(0)}{G(d)}.$$

For $d = 1$, the same conclusion follows by letting $t \rightarrow \infty$.

If $f(0) = 0$, write $f(t) = t^r g(t)$ with $g(0) > 0$. For $r = 1$,

$$f'(0) = g(0) \geq G(d) \inf_{t>0} \frac{f(t)}{t},$$

because the infimum is at most the limit $g(0)$ as $t \downarrow 0$. If $r \geq 2$, then both $f'(0)$ and the infimum in (3.2) are zero. $\square$

Theorem 3.3 (Derivative-capacity inequality). *Let p be a homogeneous real-stable polynomial of degree m in m variables, with nonnegative coefficients. Put*

$$d = \deg_{x_m} p$$

and

$$q(x_1, \dots, x_{m-1}) = \left. \frac{\partial p}{\partial x_m} \right|_{x_m=0}.$$

If $d \geq 1$, then

$$\text{Cap}(q) \geq G(d) \text{Cap}(p). \quad (3.3)$$

Proof. Fix $x' = (x_1, \dots, x_{m-1}) > 0$ and set

$$f(t) = p(x', t).$$

By Lemma 3.1, f is stable as a univariate real polynomial. Hence all its zeros are real: a nonreal zero below the real axis would have a conjugate zero above it. Nonnegative coefficients rule out positive zeros. Thus Lemma 3.2 applies.

For every $t > 0$, the definition of capacity gives

$$\frac{f(t)}{t} \geq \text{Cap}(p) x_1 \cdots x_{m-1}.$$

Therefore

$$q(x') = f'(0) \geq G(d) \text{Cap}(p) x_1 \cdots x_{m-1}.$$

Divide by $x_1 \cdots x_{m-1}$ and take the infimum over $x' > 0$. $\square$

Corollary 3.4 (Square-free coefficient extraction). *Let p be homogeneous, real stable, of degree m in m variables, and have nonnegative coefficients. Then the coefficient of $x_1 \cdots x_m$ is at least*

$$v_m \text{Cap}(p).$$

Proof. If $\text{Cap}(p) = 0$, the assertion is immediate. Otherwise differentiate successively in $x_m, x_{m-1}, \dots, x_1$, setting each differentiated variable to zero. Theorem 3.3 keeps every intermediate capacity positive. At the stage with j variables remaining, the relevant degree is at most j and at least one; if it were zero, the polynomial would be independent of a denominator variable and would have capacity zero. Since G is decreasing, Theorem 3.3 contributes at least $G(j)$. Equation (3.1) completes the proof. $\square$

3.2 The support-sensitive permanent inequality

Theorem 3.5 (Support-sensitive permanent bound). *Let $B = (b_{ij})$ be an $n \times n$ doubly stochastic matrix. Order its columns arbitrarily, and let d_j be the number of positive entries in the j th ordered column. Then*

$$\text{per}(B) \geq \prod_{j=1}^n G(\min\{j, d_j\}). \quad (3.4)$$

Proof. Consider

$$P_B(x_1, \dots, x_n) = \prod_{i=1}^n \left(\sum_{j=1}^n b_{ij} x_j \right).$$

Every linear factor has positive imaginary part when all variables lie in $\mathbb{H}$, so P_B is real stable. Weighted AM–GM gives, for $x_j > 0$,

$$\sum_j b_{ij} x_j \geq \prod_j x_j^{b_{ij}}.$$

Multiplying over the rows and using the column sums,

$$P_B(x) \geq x_1 \cdots x_n.$$

Equality holds at $x_1 = \cdots = x_n = 1$, and therefore

$$\text{Cap}(P_B) = 1. \quad (3.5)$$

Set $p_n = P_B$ and recursively

$$p_{j-1} = \frac{\partial p_j}{\partial x_j} \Big|_{x_j=0} \quad (j = n, n-1, \dots, 1).$$

At stage j , the polynomial has degree j in j variables. Differentiation in other variables cannot increase its degree in x_j , so

$$\deg_{x_j} p_j \leq \min\{j, d_j\}.$$

Using the monotonicity of G , apply Theorem 3.3 at every stage and using (3.5),

$$p_0 \geq \prod_{j=1}^n G(\min\{j, d_j\}).$$

The scalar p_0 is the coefficient of $x_1 \cdots x_n$ in P_B , namely $\text{per}(B)$. □

3.3 Zero-pattern and entropy consequences

Corollary 3.6 (One zero). *If a doubly stochastic B has at least one zero entry, then*

$$\text{per}(B) \geq \kappa_n, \quad \kappa_n := v_{n-1} G(n-1) = \frac{(n-2)!(n-2)^{n-2}}{(n-1)^{2n-4}}. \quad (3.6)$$

Moreover $\kappa_n > \gamma_n$ for every $n \geq 3$.

Proof. Put a column containing a zero last in Theorem 3.5. Its support size is at most $n - 1$, giving (3.6).

For $m = n - 1 \geq 2$,

$$\frac{\kappa_n}{\gamma_n} = \left(1 + \frac{1}{m}\right)^m \left(1 - \frac{1}{m}\right)^{m-1} = \left(1 + \frac{1}{m}\right) \left(1 - \frac{1}{m^2}\right)^{m-1}.$$

Bernoulli's inequality yields

$$\left(1 - \frac{1}{m^2}\right)^{m-1} \geq 1 - \frac{m-1}{m^2} = 1 - \frac{1}{m} + \frac{1}{m^2}.$$

Multiplication by $1 + 1/m$ gives $1 + 1/m^3 > 1$. $\square$

Theorem 3.7 (Sharp rectangular-zero-block bound). *Let B be doubly stochastic and suppose that, after row and column permutations, it contains an $a \times b$ zero block, where*

$$a, b \geq 1, \quad p := n - a - b \geq 1.$$

Then

$$\text{per}(B) \geq L_{n,a,b}^\# := \frac{v_{n-a}v_{n-b}}{v_p}. \quad (3.7)$$

The bound is sharp.

Proof. Let C be the b columns in the zero block and let R be its a rows. In P_B , every variable indexed by C occurs only in the $n - a$ row factors indexed by R^c .

Differentiate successively in the b variables indexed by C , setting each differentiated variable to zero. After k such differentiations, every surviving product-rule term has differentiated k distinct row factors: each row factor is linear and cannot be differentiated twice. Consequently, the degree of the next special variable is at most $n - a - k$. Theorem 3.3 therefore supplies the factor

$$\prod_{k=0}^{b-1} G(n - a - k) = \prod_{j=p+1}^{n-a} G(j) = \frac{v_{n-a}}{v_p}.$$

A homogeneous stable polynomial of degree $n - b$ in $n - b$ variables remains. Corollary 3.4 supplies the additional factor v_{n-b} . Since $\text{Cap}(P_B) = 1$, (3.7) follows.

For sharpness, consider

$$B_0 = \begin{pmatrix} 0_{a \times b} & \frac{1}{n-b} J_{a,n-b} \\ \frac{1}{n-a} J_{n-a,b} & \frac{p}{(n-a)(n-b)} J_{n-a,n-b} \end{pmatrix}.$$

It is doubly stochastic. Every contributing permutation uses a upper-right entries, b lower-left entries, and p lower-right entries. Their number is

$$\frac{(n-b)!}{p!} \cdot \frac{(n-a)!}{p!} \cdot p! = \frac{(n-a)!(n-b)!}{p!}.$$

Multiplying by the common weight gives

$$\text{per}(B_0) = \frac{(n-a)!(n-b)!}{p!} \frac{p^p}{(n-a)^{n-a}(n-b)^{n-b}} = \frac{v_{n-a}v_{n-b}}{v_p}.$$

$\square$

For the middle dimensions it is convenient to record a weaker consequence with a simpler form.

Corollary 3.8. *Suppose $1 \leq a \leq b$ and a doubly stochastic B contains an $a \times b$ zero block. Then*

$$\text{per}(B) \geq L_{n,a,b} := v_{n-a} G(n-b)^a. \quad (3.8)$$

In particular, a 2×2 zero block gives

$$\text{per}(B) \geq v_{n-2} G(n-2)^2. \quad (3.9)$$

Proof. Transpose first. The transposed zero block has a columns, each supported on at most $n-b$ rows. Put those a columns last in Theorem 3.5; the first $n-a$ factors give v_{n-a} and the last a factors give $G(n-b)^a$. $\square$

Theorem 3.9 (Column-entropy permanent bound). *Let B be doubly stochastic, and let $b = (b_1, \dots, b_n)$ be any one of its columns. Put*

$$s(b) = \sum_{i=1}^n \left(b_i - \frac{1}{n} \right)^2.$$

Then

$$\text{per}(B) \geq \gamma_n \exp \left(\frac{s(b)}{2} \right). \quad (3.10)$$

Proof. It is enough first to consider $0 < b_i < 1$; the general case follows by continuity. Take the selected column to be the last one and set

$$L_i(x_1, \dots, x_{n-1}) = \sum_{j < n} b_{ij} x_j.$$

The derivative polynomial is

$$q = \left. \frac{\partial P_B}{\partial x_n} \right|_{x_n=0} = \sum_i b_i \prod_{k \neq i} L_k.$$

Weighted AM–GM, with weights b_i , gives

$$q \geq \prod_i L_i^{1-b_i}.$$

A second weighted AM–GM inequality gives

$$L_i \geq (1-b_i) \prod_{j < n} x_j^{b_{ij}/(1-b_i)}.$$

Therefore

$$q(x) \geq \left(\prod_i (1-b_i)^{1-b_i} \right) x_1 \cdots x_{n-1},$$

so

$$\text{Cap}(q) \geq \prod_i (1-b_i)^{1-b_i}.$$

Corollary 3.4 yields

$$\text{per}(B) \geq v_{n-1} \prod_i (1 - b_i)^{1-b_i}. \quad (3.11)$$

Let $h(t) = (1 - t) \log(1 - t)$. Since $h''(t) = 1/(1 - t) \geq 1$ on $[0, 1]$,

$$h(t) \geq h(1/n) + h'(1/n)(t - 1/n) + \frac{1}{2}(t - 1/n)^2.$$

Summing at $t = b_i$, the linear terms cancel. Hence

$$\sum_i h(b_i) \geq nh(1/n) + \frac{1}{2}s(b).$$

Now $\exp(nh(1/n)) = G(n)$ and $v_{n-1}G(n) = \gamma_n$. Substitution into (3.11) proves (3.10). $\square$

3.4 Complement entropy and core-marginal refinements

The column-entropy estimate retains one column of a doubly stochastic matrix. The lower-dimensional boundary arguments also require the dual information carried by all complementary minors of a column and, in Dimension 6, by the marginal distributions of a whole Hall core.

For a probability vector $q = (q_1, \dots, q_m)$, put

$$\Psi_m(q) = \sum_{i=1}^m (1 - q_i) \log(1 - q_i) - m \left(1 - \frac{1}{m}\right) \log \left(1 - \frac{1}{m}\right), \quad (3.12)$$

with the convention $0 \log 0 = 0$, and write

$$s_m(q) = \sum_{i=1}^m \left(q_i - \frac{1}{m}\right)^2.$$

Lemma 3.10 (Quadratic complement-entropy bounds). *For every probability vector q of length $m \geq 2$,*

$$\Psi_m(q) \geq \left(\frac{1}{2} + \frac{1}{3m}\right) s_m(q) \geq \frac{1}{2} s_m(q). \quad (3.13)$$

Proof. Let $h(x) = (1 - x) \log(1 - x)$. Since

$$h''(x) = \frac{1}{1 - x} \geq 1 + x \quad (0 \leq x < 1),$$

For $q_i < 1$, Taylor's formula with integral remainder about $1/m$ gives, for $d_i = q_i - 1/m$,

$$h(q_i) \geq h(1/m) + h'(1/m)d_i + \frac{1 + 1/m}{2}d_i^2 + \frac{1}{6}d_i^3.$$

Because $d_i \geq -1/m$, one has $d_i^3 \geq -m^{-1}d_i^2$. Summation and $\sum_i d_i = 0$ prove the first inequality. Boundary coordinates $q_i = 1$ follow by continuity. The second inequality is immediate. $\square$

Theorem 3.11 (Complementary-minor entropy). *Let M be an $m \times m$ doubly stochastic matrix, let $q = (q_1, \dots, q_m)$ be one of its columns, and let*

$$d_i = \text{per } M(i \mid j_0)$$

be the permanent of the submatrix obtained by deleting row i and the selected column j_0 . With the convention that a factor of exponent zero is omitted,

$$\prod_{i=1}^m d_i^{q_i} \geq v_m \exp(\Psi_m(q)). \quad (3.14)$$

Consequently,

$$\prod_{i=1}^m d_i^{q_i} \geq v_m \exp \left\{ \left(\frac{1}{2} + \frac{1}{3m} \right) s_m(q) \right\} \geq v_m \exp \left(\frac{s_m(q)}{2} \right). \quad (3.15)$$

Proof. It is enough first to treat the positive case. For a general M , apply the argument to $M_\varepsilon = (1 - \varepsilon)M + \varepsilon U_m$ and let $\varepsilon \downarrow 0$. Each complementary minor is then a positive polynomial in ε ; if its limiting exponent is zero, the corresponding logarithmic contribution is $O(\varepsilon |\log \varepsilon|)$ and tends to zero. Thus the asserted product passes to the limit.

Take the selected column to be the last one and write

$$L_i(x_1, \dots, x_{m-1}) = \sum_{j < m} m_{ij} x_j.$$

For arbitrary $t_1, \dots, t_m > 0$, set

$$q_t(x) = \sum_{i=1}^m q_i t_i \prod_{k \neq i} L_k(x).$$

This is homogeneous, real stable, and of degree $m - 1$, because

$$q_t = \frac{\partial}{\partial z} \prod_{i=1}^m (L_i(x) + q_i t_i z) \Big|_{z=0}.$$

Weighted AM–GM, first with weights q_i and then within each row, gives

$$\begin{aligned} q_t &= \left(\prod_i L_i \right) \sum_i q_i \frac{t_i}{L_i} \\ &\geq \left(\prod_i t_i^{q_i} \right) \prod_i L_i^{1-q_i} \\ &\geq \left(\prod_i t_i^{q_i} \right) \left(\prod_i (1 - q_i)^{1-q_i} \right) x_1 \cdots x_{m-1}. \end{aligned}$$

Thus

$$\text{Cap}(q_t) \geq \prod_i t_i^{q_i} \prod_i (1 - q_i)^{1-q_i}.$$

The coefficient of $x_1 \cdots x_{m-1}$ in q_t is $\sum_i q_i t_i d_i$, so Corollary 3.4 yields

$$\sum_i q_i t_i d_i \geq v_{m-1} \prod_i t_i^{q_i} \prod_i (1 - q_i)^{1-q_i}.$$

This inequality implies $d_i > 0$ whenever $q_i > 0$. Put $t_i = d_i^{-1}$ on the support of q . Since $\sum_i q_i = 1$,

$$\prod_i d_i^{q_i} \geq v_{m-1} \prod_i (1 - q_i)^{1-q_i} = v_m \exp(\Psi_m(q)),$$

where $v_{m-1}G(m) = v_m$. Lemma 3.10 gives the stated consequences. $\square$

The case in which a critical zero block leaves only one unit of mass in the opposite corner will be called a *single-bridge* configuration.

Theorem 3.12 (Single-bridge zero-block entropy). *Let B be an $n \times n$ doubly stochastic matrix with a zero block $B[R, C] = 0$, where $|R| = a$, $|C| = b$, and $a + b = n - 1$. Put $I = R^c$, $J = C^c$, $m = b + 1$, and $\ell = a + 1$. The core $X = B[I, J]$ has total mass one. Let ξ and η be its row- and column-sum vectors, and put*

$$s_I = s_m(\xi), \quad s_J = s_\ell(\eta).$$

Then

$$\text{per}(B) \geq v_m v_\ell \exp(\Psi_m(\xi) + \Psi_\ell(\eta)). \quad (3.16)$$

In particular, with $c_q = \frac{1}{2} + \frac{1}{3q}$,

$$\text{per}(B) \geq v_m v_\ell \exp(c_m s_I + c_\ell s_J) \geq v_m v_\ell \exp\left(\frac{s_I + s_J}{2}\right). \quad (3.17)$$

The constant $v_m v_\ell$ is $L_{n,a,b}^\sharp$.

Proof. Every perfect matching uses exactly one edge of X . For $i \in I$ and $j \in J$, put

$$e_i = \text{per } B[I \setminus \{i\}, C], \quad d_j = \text{per } B[R, J \setminus \{j\}].$$

Expansion according to the unique core edge gives

$$\text{per}(B) = \sum_{i \in I, j \in J} x_{ij} e_i d_j.$$

The x_{ij} sum to one, so weighted AM–GM implies

$$\text{per}(B) \geq \prod_{i \in I} e_i^{\xi_i} \prod_{j \in J} d_j^{\eta_j}.$$

Adjoin ξ as a last column to $B[I, C]$. The resulting $m \times m$ matrix is doubly stochastic and its complementary minors are the e_i . Similarly, adjoining η to the transpose of $B[R, J]$ gives an $\ell \times \ell$ doubly stochastic matrix whose complementary minors are the d_j . Apply Theorem 3.11 on both sides, and then Lemma 3.10. $\square$

The next estimate retains one supported column inside an arbitrary rectangular zero block.

Theorem 3.13 (Distinguished zero-block column). *Let B be an $n \times n$ doubly stochastic matrix containing an $a \times b$ zero block $B[R, C] = 0$, where*

$$a, b \geq 1, \quad p = n - a - b \geq 1.$$

Put $I = R^c$ and $m = n - a$. Select a column $j_0 \in C$, and let $q = (b_{ij_0})_{i \in I}$ be its probability vector on I . Then

$$\text{per}(B) \geq L_{n,a,b}^\# \exp(\Psi_m(q)). \quad (3.18)$$

Consequently,

$$\text{per}(B) \geq L_{n,a,b}^\# \exp\left(\frac{s_m(q)}{2}\right).$$

The transposed assertion holds for a selected row in R .

Proof. First assume that B is positive on every position outside the prescribed zero block. Differentiate the product polynomial P_B first in the selected column variable and set that variable to zero. With

$$L_i(x) = \sum_{j \neq j_0} b_{ij} x_j,$$

the resulting homogeneous real-stable polynomial is

$$Q(x) = \sum_{i \in I} q_i \prod_{k \neq i} L_k(x).$$

Exactly as in the column-entropy proof, two weighted AM–GM inequalities give

$$\text{Cap}(Q) \geq \prod_{i \in I} (1 - q_i)^{1 - q_i}.$$

Differentiate next in the remaining $b - 1$ variables indexed by C . Their successive degrees are at most

$$m - 1, m - 2, \dots, p + 1,$$

so Theorem 3.3 supplies v_{m-1}/v_p . Square-free coefficient extraction in the remaining $n - b$ variables supplies v_{n-b} . Therefore

$$\text{per}(B) \geq \frac{v_{m-1} v_{n-b}}{v_p} \prod_{i \in I} (1 - q_i)^{1 - q_i}.$$

Since $v_{m-1} G(m) = v_m$, this is exactly (3.18).

For a general B , mix it with the block-constant extremizer from Theorem 3.7. This preserves the zero block and is positive on every allowed entry. Let the mixing parameter tend to zero; all quantities in the asserted inequality are continuous under the convention $0 \log 0 = 0$. $\square$

We shall use the distinguished-column estimate through the following symmetric consequence. Let $B[R, C] = 0$ be as above, put

$$I = R^c, \quad J = C^c, \quad m = n - a, \quad \ell = n - b,$$

and note that the core $B[I, J]$ has total mass p . Define its normalized row and column marginals by

$$\xi_i = \frac{1}{p} \sum_{j \in J} b_{ij}, \quad \eta_j = \frac{1}{p} \sum_{i \in I} b_{ij},$$

and set $S_\xi = s_m(\xi)$ and $S_\eta = s_\ell(\eta)$.

Proposition 3.14 (Core-marginal zero-block entropy). *Let $L = L_{n,a,b}^\sharp$. If $p \geq 2$, then*

$$\text{per}(B) \geq L(1 + c_{a,b}(S_\xi + S_\eta)), \quad c_{a,b} = \frac{p^2}{2(a^2 + b^2)}. \quad (3.19)$$

If $p = 1$, then

$$\text{per}(B) \geq L(1 + c_m S_\xi + c_\ell S_\eta), \quad c_q = \frac{1}{2} + \frac{1}{3q}. \quad (3.20)$$

Proof. Assume first that $p \geq 2$. For each of the b columns indexed by C , view its entries on I as a probability vector. Their average is

$$\bar{q}_i = \frac{1 - p\xi_i}{b},$$

and hence

$$\sum_{i \in I} \left(\bar{q}_i - \frac{1}{m} \right)^2 = \frac{p^2}{b^2} S_\xi.$$

Theorem 3.13, convexity of Ψ_m , and Lemma 3.10 give

$$\log \frac{\text{per}(B)}{L} \geq \max_{j \in C} \Psi_m(q^{(j)}) \geq \frac{1}{b} \sum_{j \in C} \Psi_m(q^{(j)}) \geq \Psi_m(\bar{q}) \geq \frac{p^2}{2b^2} S_\xi.$$

Transposition gives

$$\log \frac{\text{per}(B)}{L} \geq \frac{p^2}{2a^2} S_\eta.$$

The maximum of these two lower bounds is at least their weighted mean with weights $b^2/(a^2 + b^2)$ and $a^2/(a^2 + b^2)$. Thus

$$\log \frac{\text{per}(B)}{L} \geq \frac{p^2}{2(a^2 + b^2)} (S_\xi + S_\eta).$$

Use $\exp x \geq 1 + x$.

If $p = 1$, apply Theorem 3.12 and then $\exp x \geq 1 + x$. □

4 Positive global maximizers are uniform

This section gives an independent proof of the interior classification. No permanent lower bound is used.

4.1 Flatness under pairwise averaging

Choose two columns u, v of a positive matrix $A \in K_n$. For $t \in \mathbb{R}$, replace them by

$$u(t) = (1 - t)u + tv, \quad v(t) = tu + (1 - t)v, \quad (4.1)$$

and leave every other column fixed. Denote the resulting matrix by $A(t)$.

Lemma 4.1 (Pairwise averaging is flat at a positive maximizer). *If A is a positive global maximizer of Φ , then*

$$\Phi(A(t)) = \Phi(A) \quad \text{for every } t \in \mathbb{R}. \quad (4.2)$$

In particular, replacing two columns by their arithmetic mean preserves the global maximum value. The same statement holds for two rows.

Proof. Because A is positive, $A(t) \in K_n$ for all t in an open interval containing zero. The row sums are unchanged along the path. The product of the column sums is a quadratic polynomial in t , and multilinearity of the permanent in the two changed columns shows that $\text{per}(A(t))$ is also quadratic. Thus

$$F(t) := \Phi(A(t))$$

has degree at most two.

Replacing t by $1 - t$ interchanges the two columns, so $F(t) = F(1 - t)$. Hence

$$F(t) = F(0) + \lambda t(1 - t)$$

for some real λ . Since $A = A(0)$ is a local maximum along a two-sided feasible interval, $F'(0) = 0$, and therefore $\lambda = 0$. The row version follows by transposition. $\square$

4.2 Strict local maximality of the uniform matrix

Lemma 4.2 (Hessian at the uniform matrix). *Let $X = (x_{ij})$ satisfy*

$$\sum_{i,j} x_{ij} = 0.$$

Put

$$u_i = \sum_j x_{ij}, \quad v_j = \sum_i x_{ij},$$

and

$$Q = \sum_{i,j} x_{ij}^2, \quad R_2 = \sum_i u_i^2, \quad C_2 = \sum_j v_j^2.$$

Then

$$\left. \frac{d^2}{dt^2} \Phi(U_n + tX) \right|_{t=0} = -(1 - c_n)(R_2 + C_2) - c_n Q, \quad (4.3)$$

where

$$c_n = \frac{(n-2)!}{n^{n-2}}.$$

Consequently, the Hessian of Φ at U_n is negative definite on the tangent hyperplane of K_n .

Proof. The row sums of $U_n + tX$ are $1 + tu_i$. Since $\sum_i u_i = 0$,

$$\left. \frac{d^2}{dt^2} \prod_i (1 + tu_i) \right|_{t=0} = -R_2.$$

The column-product contribution is similarly $-C_2$.

For the permanent, direct differentiation gives

$$\left. \frac{d^2}{dt^2} \text{per}(U_n + tX) \right|_{t=0} = c_n \sum_{\substack{i \neq k \\ j \neq \ell}} x_{ij} x_{k\ell}. \quad (4.4)$$

Indeed, for fixed ordered pairs $i \neq k$ and $j \neq \ell$, there are $(n-2)!$ permutations satisfying $\sigma(i) = j$ and $\sigma(k) = \ell$, while the remaining $n-2$ entries contribute $n^{-(n-2)}$.

Expanding the identity

$$0 = \left(\sum_{i,j} x_{ij} \right)^2$$

according to whether two entries have the same row, the same column, or neither, gives

$$\sum_{\substack{i \neq k \\ j \neq \ell}} x_{ij} x_{k\ell} = Q - R_2 - C_2. \quad (4.5)$$

Substituting (4.5) into (4.4) proves (4.3). Since $0 < c_n \leq 1$ and $Q > 0$ for $X \neq 0$, the right side is strictly negative. $\square$

Corollary 4.3. *The matrix U_n is a strict local maximizer of Φ on K_n .*

Proof. The tangent space of the affine hull of K_n is finite-dimensional. By Lemma 4.2, the quadratic form given by the Hessian is uniformly negative on its unit sphere. Taylor's formula then gives a relative neighborhood $\mathcal{N}$ of U_n in K_n such that

$$A \in \mathcal{N}, A \neq U_n \implies \Phi(A) < \Phi(U_n).$$

$\square$

4.3 Averaging convergence and the interior classification

Lemma 4.4 (Convergence of cyclic pairwise averaging). *Let $c_1, \dots, c_n$ be vectors, and repeatedly replace pairs by their arithmetic mean according to a cyclic schedule containing every unordered pair. Then all vectors converge to their common initial mean.*

Proof. Let $\bar{c} = n^{-1} \sum_j c_j$, which is invariant, and define

$$V = \sum_j \|c_j - \bar{c}\|^2.$$

Averaging c_s, c_t decreases V by

$$\frac{1}{2} \|c_s - c_t\|^2. \quad (4.6)$$

Thus V decreases to a limit $L \geq 0$.

Assume $L > 0$. Take a convergent subsequence of the states at the beginnings of complete cycles. The total decrease during each selected cycle tends to zero. Since that decrease is a finite sum of the nonnegative quantities in (4.6), every pair difference at the moment it is averaged tends to zero. Passing to the limiting cycle, every averaging operation makes no change. Since every unordered pair occurs, all limiting vectors are equal, forcing $V = 0$, a contradiction. Hence $L = 0$. $\square$

Theorem 4.5 (Positive-maximizer theorem). *For every $n \geq 2$, every positive global maximizer of Φ on K_n is U_n .*

Proof. Let A be a positive global maximizer. Repeatedly average pairs of columns in a cyclic schedule containing every unordered pair. By Lemma 4.1, every matrix in the sequence is a positive global maximizer with the same value as A . By Lemma 4.4, the sequence converges to a positive matrix C with all columns equal. Continuity shows that C is also a global maximizer.

Now cyclically average pairs of rows of C . Every matrix remains a positive global maximizer, and the sequence converges to U_n : the equal columns of C each have sum one, and row averaging preserves every column sum. Therefore

$$\Phi(U_n) = \max_{K_n} \Phi. \quad (4.7)$$

Let Y_m be the row-averaging sequence converging to U_n . Corollary 4.3 implies that $Y_m = U_n$ for all sufficiently large m , because every Y_m has the value $\Phi(U_n)$.

We claim that one pair-averaging operation cannot map a distinct positive global maximizer X to U_n . If it did, the path (4.1) would have constant value by Lemma 4.1, and points with $t \neq 1/2$ arbitrarily close to $1/2$ would be distinct from U_n , arbitrarily close to it, and have value $\Phi(U_n)$. This contradicts strict local maximality. Hence the predecessor of U_n under such an averaging operation is also U_n .

Applying this observation backward through the finite row-averaging segment gives $C = U_n$. The column-averaging sequence converges to U_n and has constant value, so strict local maximality again forces some finite column-averaging iterate to equal U_n . Applying the same backward argument gives $A = U_n$. $\square$

Remark 4.6. Theorem 4.5 reproduces the positive-support conclusion historically due to Hwang, but its proof here is independent of [5].

5 Joint deficits, subset sums, and Hall dilation

Let A be a global maximizer of Φ on K_n . Since U_n is feasible,

$$\Phi(A) \geq \Phi(U_n) = 2 - \gamma_n. \quad (5.1)$$

Write

$$\prod_i r_i = 1 - \rho, \quad \prod_j c_j = 1 - \sigma, \quad \text{per}(A) = \gamma_n - \delta. \quad (5.2)$$

Lemma 5.1 (Joint deficit budget). *The quantities in (5.2) satisfy*

$$0 \leq \delta \leq \gamma_n, \quad \rho \geq 0, \quad \sigma \geq 0, \quad \rho + \sigma \leq \delta. \quad (5.3)$$

In particular, all row sums and column sums of A are positive.

Proof. The row sums and column sums each have arithmetic mean one, so AM–GM gives

$$\prod_i r_i \leq 1, \quad \prod_j c_j \leq 1.$$

Thus $\rho, \sigma \geq 0$. Also

$$\text{per}(A) = \prod_i r_i + \prod_j c_j - \Phi(A) \leq 2 - (2 - \gamma_n) = \gamma_n,$$

which gives $0 \leq \delta \leq \gamma_n$. Substitution into (5.1) yields

$$2 - \rho - \sigma - \gamma_n + \delta \geq 2 - \gamma_n,$$

and hence $\rho + \sigma \leq \delta$. If a row or column sum vanished, one of ρ, σ would equal one, contradicting $\delta \leq \gamma_n < 1$. $\square$

5.1 Hall domination

For a nonnegative matrix S and index sets $I, J \subseteq [n]$, write

$$s_S(I, J) = \sum_{i \in I, j \in J} s_{ij}.$$

Lemma 5.2 (Hall domination). *A nonnegative $n \times n$ matrix S dominates a doubly stochastic matrix entrywise if and only if*

$$s_S(I, J) \geq |I| + |J| - n \quad (5.4)$$

for every $I, J \subseteq [n]$ with $|I| + |J| > n$.

If equality holds in (5.4) for some I, J , then every doubly stochastic $B \leq S$ satisfies

$$B[I^c, J^c] = 0. \quad (5.5)$$

Proof. Construct a flow network with source-to-row capacities one, row-to-column capacities s_{ij} , and column-to-sink capacities one. A flow of value n is exactly a doubly stochastic matrix $B \leq S$. A cut having row set I and column set K on the source side has capacity

$$n - |I| + s_S(I, K^c) + |K|.$$

By max-flow/min-cut, every cut has capacity at least n exactly when (5.4) holds, after writing $J = K^c$.

Let

$$p = |I| + |J| - n > 0.$$

Every doubly stochastic B satisfies

$$s_B(I, J) - s_B(I^c, J^c) = p. \quad (5.6)$$

If $s_S(I, J) = p$ and $B \leq S$, then $s_B(I, J) \leq p$. Equation (5.6) and nonnegativity give the reverse inequality, so equality holds and $s_B(I^c, J^c) = 0$. $\square$

5.2 Relative-entropy subset estimates

For $0 < p, q < 1$, let

$$D(p\|q) = p \log \frac{p}{q} + (1-p) \log \frac{1-p}{1-q}.$$

We shall use the integral identity

$$D(p\|q) = \int_p^q \frac{u-p}{u(1-u)} du. \quad (5.7)$$

It follows by differentiating with respect to q . Since $u(1-u) \leq 1/4$, (5.7) also gives the binary Pinsker bound

$$D(p\|q) \geq 2(p-q)^2. \quad (5.8)$$

For $7 \leq n \leq 14$, define η_n by

n	7	8	9	10	11	12	13	14
η_n	1/47	1/80	1/135	1/230	1/350	1/650	1/1100	1/1800.

(5.9)

For $0 \leq k \leq n-1$, define

$$\alpha_{n,0} = 0$$

and, for $k \geq 1$,

$$\alpha_{n,k} = \begin{cases} 2n \left(\frac{k}{n} + \eta_n \right) \left(1 - \frac{k}{n} - \eta_n \right), & 2k < n, \\ \frac{2k(n-k)}{n}, & 2k \geq n. \end{cases} \quad (5.10)$$

The exact arithmetic in Appendix A verifies

$$\eta_n^2 > \frac{\gamma_n}{2n(1-\gamma_n)} \quad (5.11)$$

and

$$\frac{\lfloor (n-1)/2 \rfloor}{n} + \eta_n < \frac{1}{2} \quad (5.12)$$

for $7 \leq n \leq 14$.

Lemma 5.3 (Subset excess, dimensions 7 through 14). *Let $7 \leq n \leq 14$. If E is a set of k rows, $1 \leq k \leq n-1$, and*

$$\sum_{i \in E} r_i = k + \varepsilon, \quad \varepsilon > 0,$$

then

$$\varepsilon^2 \leq \frac{\alpha_{n,k}\rho}{1-\delta}. \quad (5.13)$$

The analogous estimate holds for column subsets, with σ in place of ρ .

Proof. AM–GM applied separately on E and E^c gives

$$1 - \rho \leq \left(1 + \frac{\varepsilon}{k} \right)^k \left(1 - \frac{\varepsilon}{n-k} \right)^{n-k}. \quad (5.14)$$

Put

$$p = \frac{k}{n}, \quad q = \frac{k + \varepsilon}{n}.$$

Taking logarithms in (5.14),

$$nD(p||q) \leq -\log(1 - \rho) \leq \frac{\rho}{1 - \rho} \leq \frac{\rho}{1 - \delta}. \quad (5.15)$$

Pinsker's bound and Lemma 5.1 imply

$$(q - p)^2 \leq \frac{\gamma_n}{2n(1 - \gamma_n)} < \eta_n^2. \quad (5.16)$$

If $p < 1/2$, then (5.12) and (5.16) imply $q < p + \eta_n < 1/2$. Hence

$$u(1 - u) \leq (p + \eta_n)(1 - p - \eta_n) \quad (p \leq u \leq q).$$

Using (5.7),

$$nD(p||q) \geq \frac{\varepsilon^2}{2n(p + \eta_n)(1 - p - \eta_n)} = \frac{\varepsilon^2}{\alpha_{n,k}}.$$

If $p \geq 1/2$, then $u(1 - u) \leq p(1 - p)$ for $u \in [p, q]$, giving the same conclusion with $\alpha_{n,k} = 2np(1 - p)$. Combine this lower bound with (5.15). $\square$

For $n = 15$, a sharper singleton constant is needed. Set

$$\alpha_{15,0} = 0, \quad \alpha_{15,1} = \frac{195}{98}, \quad \alpha_{15,k} = \frac{15}{2} \quad (2 \leq k \leq 14). \quad (5.17)$$

Lemma 5.4 (Subset excess in dimension 15). *The conclusion of Lemma 5.3 holds for $n = 15$ with the constants in (5.17).*

Proof. The bound $\alpha_{15,k} = 15/2$ follows from (5.8) exactly as in the preceding proof.

Suppose $k = 1$. The exact comparison $\gamma_{15} < 1/300000$, verified in Appendix A, gives

$$\varepsilon^2 \leq \frac{15\gamma_{15}}{2(1 - \gamma_{15})} < \frac{15}{2 \cdot 299999} < \frac{1}{196}.$$

Thus $0 < \varepsilon < 1/14$, and with $p = 1/15$ and $q = (1 + \varepsilon)/15$,

$$\frac{1}{15} < q < \frac{1}{14}.$$

On $[1/15, q]$,

$$u(1 - u) \leq \frac{13}{196}.$$

Equation (5.7) therefore gives

$$15D\left(\frac{1}{15}||q\right) \geq \frac{98}{195}\varepsilon^2.$$

Combining this with (5.15) proves

$$\varepsilon^2 \leq \frac{195}{98} \frac{\rho}{1 - \delta}.$$

$\square$

5.3 The critical Hall deficit

Define

$$\tau = \max \left\{ 0, \max_{\substack{I, J \subseteq [n] \\ |I| + |J| > n}} \left(1 - \frac{s_A(I, J)}{|I| + |J| - n} \right) \right\}. \quad (5.18)$$

Assume $\tau > 0$, and choose a critical pair I, J . Put

$$a = |I^c|, \quad b = |J^c|, \quad p = n - a - b \geq 1. \quad (5.19)$$

Lemma 5.5 (Hall-deficit estimate). *With the constants $\alpha_{n,k}$ defined above,*

$$\tau^2 \leq c_{n,a,b} \frac{\delta}{1 - \delta}, \quad c_{n,a,b} = \frac{\alpha_{n,a} + \alpha_{n,b}}{p^2}. \quad (5.20)$$

Whenever the right side is less than one, the matrix

$$S = \frac{A}{1 - \tau} \quad (5.21)$$

dominates a doubly stochastic matrix B , and every such B satisfies

$$B[I^c, J^c] = 0. \quad (5.22)$$

Every zero entry of A is also a zero entry of B .

Proof. Let

$$e = \sum_{i \in I^c} r_i - a, \quad f = \sum_{j \in J^c} c_j - b, \quad x = s_A(I^c, J^c) \geq 0.$$

Inclusion–exclusion gives

$$s_A(I, J) = p - e - f + x.$$

Criticality therefore implies

$$\tau p = e + f - x \leq e_+ + f_+. \quad (5.23)$$

By the subset estimates,

$$e_+^2 \leq \frac{\alpha_{n,a}\rho}{1 - \delta}, \quad f_+^2 \leq \frac{\alpha_{n,b}\sigma}{1 - \delta}.$$

Cauchy–Schwarz and (5.3) give

$$(e_+ + f_+)^2 \leq \frac{(\alpha_{n,a} + \alpha_{n,b})(\rho + \sigma)}{1 - \delta} \leq \frac{(\alpha_{n,a} + \alpha_{n,b})\delta}{1 - \delta}.$$

Together with (5.23), this proves (5.20).

By the definition of τ , the matrix S satisfies every Hall inequality (5.4). Lemma 5.2 gives a doubly stochastic $B \leq S$. Equality holds for the critical pair, so (5.22) follows. Finally, S has every zero of A , and $B \leq S$. $\square$

If $\tau = 0$, Lemma 5.2 gives a doubly stochastic $B \leq A$. Since both matrices have total sum n , necessarily $B = A$. Thus a boundary maximizer with $\tau = 0$ would itself be doubly stochastic with a zero, and Corollary 3.6 would give

$$\text{per}(A) \geq \kappa_n > \gamma_n,$$

contrary to Lemma 5.1. Hence every boundary maximizer must have $\tau > 0$.

5.4 Scalar exclusion lemmas

Lemma 5.6 (First-order scalar criterion). *Suppose*

$$\text{per}(B) \geq L, \quad \tau^2 \leq c \frac{\delta}{1-\delta}, \quad 0 \leq \delta \leq \gamma_n, \quad 0 \leq \tau < 1.$$

Then

$$L(1-\tau)^n - (\gamma_n - \delta) \geq M_n^{(1)}(L, c), \quad (5.24)$$

where

$$M_n^{(1)}(L, c) = L - \gamma_n - \frac{n^2 c L^2}{4(1-\gamma_n)}. \quad (5.25)$$

In particular, $M_n^{(1)}(L, c) > 0$ contradicts

$$\text{per}(A) = \gamma_n - \delta \geq L(1-\tau)^n.$$

Proof. Bernoulli's inequality gives $(1-\tau)^n \geq 1-n\tau$. Hence

$$\begin{aligned} L(1-\tau)^n - (\gamma_n - \delta) &\geq L - \gamma_n + \delta - nL\tau \\ &\geq L - \gamma_n + \delta - \frac{nL\sqrt{c}}{\sqrt{1-\gamma_n}}\sqrt{\delta}. \end{aligned}$$

Complete the square in $\sqrt{\delta}$. □

Lemma 5.7 (Second-order scalar criterion). *Suppose, in addition, that $0 \leq \tau \leq T$ and*

$$(1-t)^n \geq 1-nt + \lambda t^2 \quad (0 \leq t \leq T). \quad (5.26)$$

Then

$$L(1-\tau)^n - (\gamma_n - \delta) \geq M_n^{(2)}(L, c), \quad (5.27)$$

where

$$M_n^{(2)}(L, c) = L - \gamma_n - \frac{n^2 L^2}{4(\lambda L + (1-\gamma_n)/c)}. \quad (5.28)$$

Proof. The Hall bound implies

$$\delta \geq \frac{1-\gamma_n}{c} \tau^2.$$

Therefore

$$L(1-\tau)^n - (\gamma_n - \delta) \geq L - \gamma_n - nL\tau + \left(\lambda L + \frac{1-\gamma_n}{c} \right) \tau^2.$$

The minimum of the quadratic on the right is (5.28). □

6 Boundary exclusion in dimensions eleven through fifteen

We now apply the common framework. Throughout this section, A denotes a hypothetical boundary global maximizer. As observed after Lemma 5.5, necessarily $\tau > 0$.

6.1 Dimensions eleven through fourteen: ordinary Hall pairs

For $11 \leq n \leq 14$, let

$$\mathcal{P}_n = \{(a, b) : 0 \leq a \leq b, a + b \leq n - 1\}.$$

Transposition permits the restriction $a \leq b$. For $(a, b) \in \mathcal{P}_n$, put

$$p = n - a - b, \quad c_{n,a,b} = \frac{\alpha_{n,a} + \alpha_{n,b}}{p^2}, \quad (6.1)$$

and

$$\Lambda_{n,a,b} = \begin{cases} \kappa_n, & a = 0, \\ v_{n-a} G(n-b)^a, & a \geq 1. \end{cases} \quad (6.2)$$

If $a = 0$, the forced Hall block is empty, but B inherits a zero from A , so Corollary 3.6 gives the first line. If $a \geq 1$, Corollary 3.8 gives the second.

The constants in (5.10) satisfy $\alpha_{n,k} \leq n/2$: in the first branch this is $2nx(1-x) \leq n/2$, and in the second it is $2k(n-k)/n \leq n/2$. Hence $c_{n,a,b} \leq n$. Moreover

$$\frac{\gamma_{n+1}}{\gamma_n} = \left(\frac{n}{n+1} \right)^n < 1,$$

so $\gamma_n \leq \gamma_{11} < 1/1000$ for $11 \leq n \leq 14$. Therefore

$$\tau^2 \leq \frac{n\gamma_n}{1 - \gamma_n} < \frac{14}{999} < 1.$$

Thus the Hall dilation is valid.

The following proposition is a finite exact-rational calculation. Its formulas are explicit in (6.1), (6.2), and (5.25); Appendix B gives a complete standard-library implementation using rational arithmetic only.

Proposition 6.1 (Exact ordinary margins). *For the indicated sets of pairs,*

$$M_n^{(1)}(\Lambda_{n,a,b}, c_{n,a,b}) > 0.$$

More precisely, the minimum values are as follows:

n	minimizing pair	exact minimum
11	(0, 9)	$\frac{1260916230510486033100687951530284366056569069}{65689050117789834023535156250000000000000000000}$
12	(0, 11)	$\frac{1296568134644485856101383130605756547126510510475}{52823701911706625804697555859966244866493587958274637824}$
13	(0, 12)	$\frac{2344634926829311765047551592181776557728012393742935099475}{64121715613888291470979410224708337217950361958912179251651280896}$
14	(0, 13)	$\frac{4746695979316977001752598428103553549748722442212199016699590578325}{277886267525954703535897576147309905022601205806915212282941432903903914488}$.

(6.3)

For $n = 11$, the pair $(0, 10)$ is excluded from this proposition and is handled separately below.

Proof. Substitute the rational definitions into (5.25) for every pair in $\mathcal{P}_n$, omitting $(0, 10)$ when $n = 11$. There are respectively 35, 42, 49, 56 cases. Clearing denominators gives positive integers in every case. The exact enumeration in Appendix B also compares the resulting fractions and returns the minima displayed in (6.3). No rounded quantity is used. $\square$

Corollary 6.2. *All critical Hall pairs in dimensions 12, 13, 14 are impossible. In dimension 11, every critical pair except $(a, b) = (0, 10)$ and its transpose is impossible.*

Proof. For a critical pair, Lemma 5.5 gives B and the bound (5.20); the appropriate permanent estimate is (6.2). Proposition 6.1 and Lemma 5.6 contradict (6.4), where

$$\text{per}(A) = (1 - \tau)^n \text{per}(S) \geq (1 - \tau)^n \text{per}(B). \quad (6.4)$$

□

6.2 The exceptional deficient column in dimension eleven

Assume $n = 11$ and $(a, b) = (0, 10)$. Then $I = [11]$, while J consists of one column, say column j , and

$$c_j = 1 - \tau. \quad (6.5)$$

The complementary set of ten columns has total sum $10 + \tau$. Lemma 5.3 gives

$$\tau^2 \leq \frac{\alpha_{11,10}\sigma}{1 - \delta}. \quad (6.6)$$

We obtain a second use of τ from stationarity.

Lemma 6.3 (Deficient-column stationarity). *Let A be a global maximizer in dimension 11. Suppose its row sums are positive and a column has sum $1 - \tau < 1$. If*

$$d = \max_i (r_i - 1)_+,$$

then

$$d \geq (1 - \gamma_{11})\tau. \quad (6.7)$$

Proof. Multiply the deficient column by a parameter t and rescale the whole matrix so that the total sum remains 11. Let

$$u = \sum_i \frac{a_{ij}}{r_i}, \quad K = \frac{\Phi(A)}{\prod_i r_i}.$$

Differentiating at $t = 1$ gives

$$\left(\prod_i r_i \right) u + \prod_j c_j - \text{per}(A) - (1 - \tau)\Phi(A) = 0.$$

Since $\prod_j c_j - \text{per}(A) = \Phi(A) - \prod_i r_i$, this simplifies to

$$u = 1 - K\tau. \quad (6.8)$$

Now put $w_i = a_{ij}/r_i$. Then $w_i \geq 0$, $\sum_i w_i = u$, and

$$(1 - \tau) - u = \sum_i w_i(r_i - 1) = (K - 1)\tau.$$

The right side is positive. Moreover $0 < u < 1$ by (6.8), so

$$(K - 1)\tau \leq d \sum_i w_i = du \leq d.$$

Finally, $\Phi(A) \geq 2 - \gamma_{11}$ and $\prod_i r_i \leq 1$, so $K \geq 2 - \gamma_{11}$. □

For a thick pair, $a, b \geq 2$, so B contains a 2×2 zero block. Thus

$$\text{per}(B) \geq \mu_2, \quad \mu_2 := v_{13}G(13)^2 = \frac{13! 12^{24}}{13^{37}}. \quad (6.17)$$

Also

$$\tau^2 \leq 15 \frac{\delta}{1 - \delta}. \quad (6.18)$$

The exact comparisons

$$\gamma_{15} < \frac{1}{300000}, \quad \gamma_{15} < \frac{29863}{10^{10}}, \quad (6.19)$$

$$\frac{29937}{10^{10}} < \mu_1 < \frac{3}{10^6}, \quad (6.20)$$

and

$$\frac{301}{10^8} < \mu_2 < \frac{4}{10^6} \quad (6.21)$$

are certified by the six positive integers listed in Appendix A.

For the thin case,

$$\frac{225}{4} \frac{465}{49} = \frac{104625}{196} < 534.$$

Equations (6.19) and (6.20) give

$$\mu_1 - \gamma_{15} > \frac{74}{10^{10}}$$

and

$$\frac{225(465/49)\mu_1^2}{4(1 - \gamma_{15})} < 534 \cdot \frac{9}{10^{12}} \cdot \frac{300000}{299999} < \frac{5}{10^9}.$$

The final strict inequality follows after clearing denominators; its integer gap is

$$58\,195\,000\,000\,000\,000 > 0.$$

Therefore

$$M_{15}^{(1)}\left(\mu_1, \frac{465}{49}\right) > \frac{74}{10^{10}} - \frac{5}{10^9} = \frac{24}{10^{10}} > 0. \quad (6.22)$$

For the thick case, $\gamma_{15} < 299/10^8$, so

$$\mu_2 - \gamma_{15} > \frac{2}{10^8}.$$

Also

$$\frac{225 \cdot 15\mu_2^2}{4(1 - \gamma_{15})} < \frac{3375}{4} \cdot \frac{16}{10^{12}} \cdot \frac{300000}{299999} < \frac{14}{10^9},$$

where the final cleared-denominator gap is

$$149\,986\,000\,000\,000\,000 > 0.$$

Hence

$$M_{15}^{(1)}(\mu_2, 15) > \frac{2}{10^8} - \frac{14}{10^9} = \frac{6}{10^9} > 0. \quad (6.23)$$

Both cases contradict Lemma 5.6. Thus dimension 15 has no boundary global maximizer.

Theorem 6.4 (Boundary exclusion for $11 \leq n \leq 15$). *For every $n \in \{11, 12, 13, 14, 15\}$, no global maximizer of Φ on K_n has a zero entry.*

Proof. If $\tau = 0$, the contradiction following Lemma 5.5 applies. If $\tau > 0$, Corollary 6.2 handles dimensions 12, 13, 14 and all ordinary pairs in dimension 11; equations (6.11)–(6.14) handle the exceptional pair. The thin–thick argument handles dimension 15. $\square$

7 Boundary exclusion in dimensions eight, nine, and ten

For $n \in \{8, 9, 10\}$, define

$$\begin{array}{c|ccc} n & 8 & 9 & 10 \\ \hline T_n & 1/7 & 1/10 & 1/10 \\ \lambda_n & 20 & 27 & 33. \end{array} \quad (7.1)$$

As before, let A be a hypothetical boundary global maximizer. The constants $\alpha_{n,k}$ satisfy $\alpha_{n,k} \leq n/2$, so Lemma 5.5 gives

$$\tau^2 \leq \frac{n\gamma_n}{1 - \gamma_n} < T_n^2. \quad (7.2)$$

The exact positive gaps in (7.2) are

$$\begin{array}{c|c} n & T_n^2 - \frac{n\gamma_n}{1 - \gamma_n} \\ \hline 8 & \frac{7277}{6407093} \\ 9 & \frac{746489}{477848900} \\ 10 & \frac{994933}{156193300}. \end{array} \quad (7.3)$$

Lemma 7.1 (Quadratic binomial bounds). *For $n \in \{8, 9, 10\}$ and $0 \leq t \leq T_n$,*

$$(1 - t)^n \geq 1 - nt + \lambda_n t^2. \quad (7.4)$$

Proof. From the binomial expansion,

$$(1 - t)^n \geq 1 - nt + \binom{n}{2} t^2 - \binom{n}{3} t^3.$$

Indeed, after the cubic term the alternating terms have decreasing absolute values on the stated intervals: for $k \geq 3$,

$$\frac{\binom{n}{k+1} t^{k+1}}{\binom{n}{k} t^k} = \frac{n - k}{k + 1} t < 1.$$

Thus the remaining tail starts positive and is nonnegative. Finally,

$$\begin{aligned} \binom{8}{2} - \binom{8}{3} \frac{1}{7} &= 20, \\ \binom{9}{2} - \binom{9}{3} \frac{1}{10} &= \frac{138}{5} > 27, \end{aligned}$$

and

$$\binom{10}{2} - \binom{10}{3} \frac{1}{10} = 33.$$

$\square$

7.1 Two-sided critical pairs

Suppose $a, b \geq 1$. By transposition, assume $a \leq b$. Put

$$p = n - a - b, \quad c_{n,a,b} = \frac{\alpha_{n,a} + \alpha_{n,b}}{p^2},$$

and use the sharp block bound

$$L_{n,a,b}^\# = \frac{v_{n-a}v_{n-b}}{v_p}.$$

Lemma 5.7 applies with $\lambda = \lambda_n$.

Proposition 7.2 (Exact two-sided margins). *For every*

$$n \in \{8, 9, 10\}, \quad 1 \leq a \leq b, \quad a + b \leq n - 1,$$

one has

$$M_n^{(2)}(L_{n,a,b}^\#, c_{n,a,b}) > 0.$$

The minimum occurs at $(a, b) = (1, 1)$ and equals

n	<i>exact minimum</i>	
8	4139711814785404757611563	
	300024713494271812137373270016	
9	3087705658660821267020074863425	
	540713498338770121894081281002569728	
10	36184754863040830496579778359572513127	
	18255668299923283084495152136002059873437500	(7.5)

Proof. There are respectively 12, 16, 20 unordered pairs. Substitute the exact rational definitions into (5.28), clear denominators, and compare the resulting positive fractions. Appendix B performs exactly this finite calculation. $\square$

Thus no two-sided critical pair can occur.

7.2 One-sided cuts and stationarity

It remains to consider a one-sided critical pair. After transposition, assume $a = 0$. Then $I = [n]$ and J is a set of

$$p = n - b$$

columns satisfying

$$\sum_{j \in J} c_j = p(1 - \tau). \quad (7.6)$$

The corresponding columns of $S = A/(1 - \tau)$ have total sum p . Since $B \leq S$ is doubly stochastic, those same p columns of B also have total sum p . Hence entrywise saturation holds:

$$B[:, J] = S[:, J]. \quad (7.7)$$

Define

$$m_i = \sum_{j \in J} b_{ij}, \quad \beta_i = \frac{m_i}{p}. \quad (7.8)$$

Then $\beta = (\beta_i)$ is a probability vector.

Lemma 7.3 (One-sided stationarity identity). *Put*

$$K = \frac{\Phi(A)}{\prod_i r_i}$$

and

$$D = 1 - \sum_i \frac{\beta_i}{r_i}.$$

Then

$$D = \frac{(K-1)\tau}{1-\tau} \geq \frac{(1-\gamma_n)\tau}{1-\tau} > 0. \quad (7.9)$$

Proof. Multiply all columns in J by a parameter t , and then multiply the whole matrix by

$$\lambda(t) = \frac{n}{n + (t-1)p(1-\tau)}$$

so that the total entry sum remains n . This is a two-sided feasible curve for t near one. Let

$$u = \sum_i \frac{\sum_{j \in J} a_{ij}}{r_i}.$$

Before the global normalization, the derivative of the row-product term is $(\prod_i r_i)u$. The product of column sums and the permanent each acquire a factor t^p , so their derivatives are $p \prod_j c_j$ and $p \operatorname{per}(A)$. The derivative of the global factor $\lambda(t)^n$ is $-p(1-\tau)$. Stationarity at $t = 1$ therefore gives

$$\left(\prod_i r_i \right) u + p \left(\prod_j c_j - \operatorname{per}(A) \right) - p(1-\tau)\Phi(A) = 0.$$

Using $\prod_j c_j - \operatorname{per}(A) = \Phi(A) - \prod_i r_i$,

$$u = p - pK\tau. \quad (7.10)$$

By saturation,

$$\sum_{j \in J} a_{ij} = (1-\tau)m_i.$$

Thus (7.10), divided by p , becomes

$$(1-\tau) \sum_i \frac{\beta_i}{r_i} = 1 - K\tau,$$

which is the asserted identity. Finally,

$$K \geq \Phi(A) \geq 2 - \gamma_n,$$

because $\prod_i r_i \leq 1$. □

7.3 Row-product energy

Define

$$d_0^2 = \frac{n\gamma_n}{2(1-\gamma_n)}. \quad (7.11)$$

For $n = 8, 9, 10$, exact arithmetic gives $d_0 < 1/10$; the positive gaps are

$$\begin{array}{c|c} n & \frac{1}{100} - d_0^2 \\ \hline 8 & \frac{4757}{13075700} \\ 9 & \frac{2762489}{477848900} \\ 10 & \frac{1278433}{156193300} \end{array} \quad (7.12)$$

Lemma 7.4 (Row-product energy). *For $n \in \{8, 9, 10\}$, put*

$$y_i = 1 - \frac{1}{r_i}.$$

Then

$$\rho > \frac{2}{5} \sum_i y_i^2. \quad (7.13)$$

Proof. First, every row sum satisfies $r_i \geq 1 - d_0$. Indeed, if $r_i = 1 - d < 1$, the complementary $n - 1$ rows have positive excess d . Lemma 5.3 and $\alpha_{n,n-1} \leq n/2$ give

$$d^2 \leq \frac{n\rho}{2(1-\delta)} \leq \frac{n\gamma_n}{2(1-\gamma_n)} = d_0^2.$$

For every $r \geq 1 - d_0$,

$$r - 1 - \log r \geq \frac{(1 - d_0)^2}{2} \left(1 - \frac{1}{r}\right)^2. \quad (7.14)$$

To see this, set $y = 1 - 1/r$. Then

$$r - 1 - \log r = \int_0^y \frac{t}{(1-t)^2} dt.$$

For $y \geq 0$, the integrand is at least t ; for $y < 0$, reverse the integral and use $(1-t)^{-2} \geq (1-d_0)^2$ on $[y, 0]$.

Since $\sum_i (r_i - 1) = 0$,

$$-\log(1 - \rho) = \sum_i (r_i - 1 - \log r_i) \geq \frac{(1 - d_0)^2}{2} \sum_i y_i^2.$$

Also

$$\rho = 1 - (1 - \rho) \geq (1 - \rho)(-\log(1 - \rho)) \geq (1 - \gamma_n)(-\log(1 - \rho)).$$

For these dimensions, $\gamma_n < 1/81$ and $d_0 < 1/10$, so

$$\frac{(1 - \gamma_n)(1 - d_0)^2}{2} > \frac{80}{81} \cdot \frac{81}{100} \cdot \frac{1}{2} = \frac{2}{5}.$$

□

Let

$$s = \sum_i \left(\beta_i - \frac{1}{n} \right)^2. \quad (7.15)$$

By the harmonic-mean inequality,

$$\sum_i \frac{1}{r_i} \geq \frac{n^2}{\sum_i r_i} = n,$$

so $\sum_i y_i \leq 0$. Therefore

$$D = \sum_i \beta_i y_i \leq \sum_i \left(\beta_i - \frac{1}{n} \right) y_i \leq \sqrt{s} \left(\sum_i y_i^2 \right)^{1/2}.$$

Since $D > 0$, one has $s > 0$. Lemma 7.4 and (7.9) imply

$$\rho > \frac{2}{5} \frac{D^2}{s} \geq \frac{2}{5} \frac{(1 - \gamma_n)^2 \tau^2}{(1 - \tau)^2 s}. \quad (7.16)$$

The p columns in J are probability vectors whose average is β . Convexity of squared Euclidean distance shows that at least one of them, say $b^{(j)}$, satisfies

$$s(b^{(j)}) \geq s.$$

Theorem 3.9 and Corollary 3.6 give

$$\text{per}(B) \geq \max\{\kappa_n, \gamma_n e^{s/2}\}. \quad (7.17)$$

Consequently,

$$\text{per}(A) \geq (1 - \tau)^n \max\{\kappa_n, \gamma_n e^{s/2}\}. \quad (7.18)$$

On the other hand, $\delta \geq \rho$, and therefore

$$\text{per}(A) = \gamma_n - \delta \leq \gamma_n - \rho. \quad (7.19)$$

For a one-sided pair, (5.20), $p \geq 1$, and $\alpha_{n,b} \leq n/2$ give

$$\tau^2 \leq \frac{n\gamma_n}{2(1 - \gamma_n)} = d_0^2 < \frac{1}{100}. \quad (7.20)$$

7.4 The two entropy regimes

Set

$$s_0 = 2 \log \frac{\kappa_n}{\gamma_n} > 0.$$

Lemma 7.5 (Small entropy). *If $s \leq s_0$, then*

$$\rho + \kappa_n(1 - \tau)^n > \gamma_n. \quad (7.21)$$

Proof. From (7.16),

$$\rho > Q_n \tau^2, \quad Q_n = \frac{2(1 - \gamma_n)^2}{5s_0}.$$

Bernoulli's inequality gives

$$\rho + \kappa_n(1 - \tau)^n - \gamma_n > \kappa_n - \gamma_n - n\kappa_n\tau + Q_n\tau^2.$$

The right side is at least

$$\kappa_n - \gamma_n - \frac{n^2\kappa_n^2}{4Q_n}.$$

Since $\log x \leq x - 1$,

$$s_0 \leq \frac{2(\kappa_n - \gamma_n)}{\gamma_n}.$$

Thus the last expression is positive whenever

$$\frac{2}{5}(1 - \gamma_n)^2\gamma_n - \frac{n^2}{2}\kappa_n^2 > 0. \quad (7.22)$$

The exact positive values of the left side are

n	value in (7.22)	
8	$\frac{165702517175104104520946768451687}{215701290564174919711661340674228224}$	
9	$\frac{5586975164420125257678146526492250330079}{16534930139652222569055999993919229392846848}$	(7.23)
10	$\frac{55944735805258396002952464523577322487640863}{404273588897859606879181224832534790039062500000}$	

□

Lemma 7.6 (Large entropy). *If $s \geq s_0$, then*

$$\rho + \gamma_n(1 - \tau)^n e^{s/2} > \gamma_n. \quad (7.24)$$

Proof. Put $q = (1 - \tau)^n$. Since $e^{s/2} \geq 1 + s/2$, equation (7.16) and AM–GM give

$$\begin{aligned} \rho + \gamma_n q e^{s/2} - \gamma_n &> \frac{2}{5} \frac{D^2}{s} + \gamma_n q \left(1 + \frac{s}{2}\right) - \gamma_n \\ &\geq D \sqrt{\frac{4}{5} \gamma_n q} + \gamma_n (q - 1). \end{aligned}$$

By (7.9), (7.20), and Bernoulli's inequality,

$$D\sqrt{q} \geq (1 - \gamma_n)\tau \left(\frac{9}{10}\right)^{n/2}, \quad q - 1 \geq -n\tau.$$

Therefore the preceding expression is at least

$$\tau \left[(1 - \gamma_n) \left(\frac{9}{10}\right)^{n/2} \sqrt{\frac{4}{5} \gamma_n} - n\gamma_n \right].$$

The bracket is positive exactly when

$$\frac{4}{5}(1 - \gamma_n)^2 \left(\frac{9}{10}\right)^n - n^2 \gamma_n > 0. \quad (7.25)$$

The exact positive values are

n	value in (7.25)	(7.26)
8	$\frac{405685268407642329}{2147483648000000000}$	
9	$\frac{17233957123121}{738112500000000}$	
10	$\frac{7399058328856214668089}{305175781250000000000000}$	

□

Equations (7.18) and (7.19) contradict Lemma 7.5 when $s \leq s_0$ and Lemma 7.6 when $s \geq s_0$. Thus one-sided critical pairs are impossible.

Theorem 7.7 (Boundary exclusion for $8 \leq n \leq 10$). *For every $n \in \{8, 9, 10\}$, no global maximizer of Φ on K_n has a zero entry.*

Proof. The case $\tau = 0$ was excluded after Lemma 5.5. For $\tau > 0$, Proposition 7.2 excludes two-sided critical pairs, and the one-sided analysis above excludes the remaining pairs. □

8 Boundary exclusion in dimension seven

Dimension seven is the first dimension in which the scalar zero-block comparison alone fails. The failure is confined to critical Hall pairs with $p = n - a - b = 1$. The complementary-minor entropy from Theorem 3.12 supplies exactly the missing information.

Throughout this section let $n = 7$, write

$$\gamma = \gamma_7 = \frac{720}{117649},$$

and let A be a hypothetical boundary global maximizer. The localization parameter is

$$\eta_7 = \frac{1}{47}.$$

The two inequalities needed in Lemma 5.3 are

$$\eta_7^2 - \frac{\gamma}{14(1 - \gamma)} = \frac{23263}{1808073127} > 0, \quad \frac{1}{2} - \frac{3}{7} - \eta_7 = \frac{33}{658} > 0. \quad (8.1)$$

Thus all subset-excess estimates of Section 5 apply in dimension seven. In particular,

$$\max_{1 \leq k \leq 6} \alpha_{7,k} = \alpha_{7,3} = \frac{53576}{15463} < \frac{7}{2}. \quad (8.2)$$

As before, the case $\tau = 0$ is impossible, so fix a critical pair and use a, b, p, e, f, x as in Lemma 5.5.

8.1 Two-sided pairs with core size at least two

Put

$$T_7 = \frac{5}{24}, \quad \lambda_7 = \frac{329}{24} = \binom{7}{2} - \binom{7}{3} T_7. \quad (8.3)$$

The common Hall estimate and (8.2) imply

$$\tau^2 \leq \frac{7\gamma}{1-\gamma} < T_7^2, \quad T_7^2 - \frac{7\gamma}{1-\gamma} = \frac{20185}{67351104} > 0. \quad (8.4)$$

The alternating binomial argument of Lemma 7.1 gives

$$(1-t)^7 \geq 1 - 7t + \lambda_7 t^2 \quad (0 \leq t \leq T_7). \quad (8.5)$$

Proposition 8.1 (Ordinary dimension-seven margins). *For every $1 \leq a \leq b$ with $p = 7 - a - b \geq 2$,*

$$M_7^{(2)} \left(L_{7,a,b}^\sharp, \frac{\alpha_{7,a} + \alpha_{7,b}}{p^2} \right) > 0,$$

where (5.28) is evaluated with $\lambda = 329/24$. There are six pairs, and the minimum is attained at $(a, b) = (1, 1)$:

$$\frac{217499955444827479495}{21713873952987016007066496} > 0. \quad (8.6)$$

Proof. This is an exact rational substitution in Lemma 5.7. The six comparisons are checked by the supplementary certificate described in Appendix B. $\square$

Proposition 8.1 excludes every two-sided critical pair with $p \geq 2$.

8.2 One-sided pairs

The one-sided stationarity identity from Lemma 7.3 remains valid. We sharpen only the numerical energy estimates. Define

$$d_7^2 = \frac{7\gamma}{2(1-\gamma)} = \frac{2520}{116929}.$$

Then

$$\frac{9}{400} - d_7^2 = \frac{44361}{46771600} > 0, \quad (8.7)$$

so every row sum and every column sum of A is greater than $17/20$.

Lemma 8.2 (Dimension-seven product energy). *With $y_i = 1 - r_i^{-1}$,*

$$\rho > \frac{1}{3} \sum_i y_i^2. \quad (8.8)$$

The transposed statement holds for the column sums.

Proof. The proof of Lemma 7.4 gives

$$\rho \geq \frac{(1-\gamma)(17/20)^2}{2} \sum_i y_i^2.$$

The coefficient exceeds $1/3$, since

$$\frac{(1-\gamma)(17/20)^2}{2} - \frac{1}{3} = \frac{7258243}{282357600} > 0.$$

□

Retain the notation $D, s, q = (1-\tau)^7$ from the one-sided analysis in Section 7. Lemmas 7.3 and 8.2 give

$$\rho > \frac{D^2}{3s} \geq \frac{(1-\gamma)^2 \tau^2}{3(1-\tau)^2 s}. \quad (8.9)$$

Also $\tau < 3/20$ by (8.7).

Proposition 8.3 (One-sided exclusion in dimension seven). *No one-sided critical pair occurs at a boundary global maximizer in dimension seven.*

Proof. Set $s_0 = 2 \log(\kappa_7/\gamma)$. If $s \leq s_0$, the proof of Lemma 7.5, with $1/3$ in place of $2/5$, reduces the contradiction to

$$\frac{1}{3}(1-\gamma)^2 \gamma - \frac{49}{2} \kappa_7^2 = \frac{22176489486262543850191055}{20672701805255574480144334848} > 0. \quad (8.10)$$

Suppose $s \geq s_0$. From (8.9), $e^{s/2} \geq 1 + s/2$, and AM–GM,

$$\rho + \gamma q e^{s/2} - \gamma > D \sqrt{\frac{2\gamma q}{3}} + \gamma(q-1). \quad (8.11)$$

The stationarity bound gives

$$D\sqrt{q} \geq (1-\gamma)\tau(1-\tau)^{5/2}.$$

Moreover, for $0 \leq t \leq 3/20$,

$$\frac{1 - (1-t)^7}{t} \leq 7 - 21t + 35t^2.$$

Consequently the right side of (8.11) is positive if

$$W(t) := \frac{2}{3}(1-\gamma)^2(1-t)^6 - \gamma(7-21t+35t^2)^2 > 0. \quad (8.12)$$

The degree-six Bernstein coefficients of W on $[0, 3/20]$ are all positive; the smallest is

$$\frac{155130361249919329}{1328763571296000000} > 0. \quad (8.13)$$

This exact coefficient calculation appears in the supplementary certificate. Thus both entropy regimes contradict the upper bound $\text{per}(A) \leq \gamma - \rho$, exactly as in Section 7. □

8.3 Single-bridge pairs: preliminary reductions

It remains to treat $p = 1$. By transposition assume $a \leq b$, so

$$(a, b) \in \{(1, 5), (2, 4), (3, 3)\}. \quad (8.14)$$

Recall

$$e = \sum_{i \in I^c} r_i - a, \quad f = \sum_{j \in J^c} c_j - b, \quad x = s_A(I^c, J^c),$$

so criticality gives

$$\tau = e + f - x. \quad (8.15)$$

If $e \leq 0$, then $\tau \leq f_+$ and the proof of Lemma 5.5 gives the sharper estimate

$$\tau^2 \leq \alpha_{7,b} \frac{\delta}{1 - \delta}.$$

If $f \leq 0$, the analogous estimate uses $\alpha_{7,a}$. Since

$$\left(\frac{3}{20}\right)^2 - \alpha_{7,3} \frac{\gamma}{1 - \gamma} = \frac{842770143}{723229250800} > 0, \quad (8.16)$$

we may use the quadratic binomial bound with

$$T = \frac{3}{20}, \quad \lambda = \binom{7}{2} - \binom{7}{3} T = \frac{63}{4}.$$

Proposition 8.4 (Sign-mixed single-bridge margins). *For the three pairs in (8.14), all six applications of Lemma 5.7 obtained from $e \leq 0$ or $f \leq 0$ have positive margin. Their minimum is*

$$\frac{165035926385}{3897251304526692} > 0, \quad (8.17)$$

attained for $(a, b) = (1, 5)$ with $e \leq 0$.

Proof. Substitute $L = L_{7,a,b}^\sharp$ and respectively $c = \alpha_{7,b}$ or $c = \alpha_{7,a}$ into (5.28), with $\lambda = 63/4$. The supplementary script checks the six exact fractions. $\square$

We may henceforth assume

$$e > 0, \quad f > 0. \quad (8.18)$$

The subset estimates and (8.2) imply

$$0 < e, f < \frac{3}{20}. \quad (8.19)$$

Put

$$t = e + f, \quad m = 7 - a = b + 1, \quad \ell = 7 - b = a + 1, \quad d = \min\{a, b\} = a.$$

Define the two group-product envelopes

$$P_a(e) = \left(1 + \frac{e}{a}\right)^a \left(1 - \frac{e}{7-a}\right)^{7-a}, \quad P_b(f) = \left(1 + \frac{f}{b}\right)^b \left(1 - \frac{f}{7-b}\right)^{7-b}. \quad (8.20)$$

If

$$P = \prod_i r_i, \quad Q = \prod_j c_j, \quad H = \text{per}(A),$$

write

$$u_r = P_a(e) - P, \quad u_c = P_b(f) - Q,$$

and set

$$D_0 = 2 - P_a(e) - P_b(f), \quad K = \gamma - D_0, \quad u = \delta - D_0. \quad (8.21)$$

Then

$$u \geq u_r + u_c \geq 0, \quad H = K - u. \quad (8.22)$$

Lemma 8.5 (Conditional product energy). *Let*

$$V_I = \sum_{i \in I} \left(\frac{1}{r_i} - \frac{1}{m} \sum_{h \in I} \frac{1}{r_h} \right)^2$$

and define V_J analogously from the columns indexed by J . Then

$$V_I \leq 3u_r, \quad V_J \leq 3u_c. \quad (8.23)$$

Proof. All row and column sums exceed $r_0 = 17/20$. Let a group of k positive numbers z_i have arithmetic mean μ . Since $\sum_i (z_i/\mu - 1) = 0$,

$$k \log \mu - \sum_i \log z_i = \sum_i \left(\frac{z_i}{\mu} - 1 - \log \frac{z_i}{\mu} \right).$$

For $z_i, \mu \geq r_0$, the integral proof of (7.14), with base point μ , gives

$$\frac{z_i}{\mu} - 1 - \log \frac{z_i}{\mu} \geq \frac{r_0^2}{2} \left(\frac{1}{z_i} - \frac{1}{\mu} \right)^2. \quad (8.24)$$

Applying this to the two row groups and using the fact that variance is minimized at the arithmetic mean of the reciprocals,

$$\log \frac{P_a(e)}{P} \geq \frac{r_0^2}{2} V_I.$$

Therefore

$$u_r = P_a(e) - P = P \left(\exp \left(\log \frac{P_a(e)}{P} \right) - 1 \right) \geq (1 - \gamma) \frac{r_0^2}{2} V_I \geq \frac{1}{3} V_I,$$

where the last inequality is the exact comparison in Lemma 8.2. The column proof is identical. $\square$

8.4 Core stationarity and the entropy squeeze

Let $q = 1 - \tau$. At a critical pair, the doubly stochastic matrix $B \leq A/q$ has $B[I^c, J^c] = 0$. Both $B[I, J]$ and $(A/q)[I, J]$ have total mass one, so the core is saturated:

$$B[I, J] = \frac{A[I, J]}{q}. \quad (8.25)$$

Let ξ and ζ be the row- and column-sum vectors of this core and put

$$s_I = \sum_{i \in I} \left(\xi_i - \frac{1}{m} \right)^2, \quad s_J = \sum_{j \in J} \left(\zeta_j - \frac{1}{\ell} \right)^2, \quad s = s_I + s_J. \quad (8.26)$$

For $k \geq 0$, let H_k be the contribution to H of permutations using exactly k entries of the block $I^c \times J^c$. Put H_0 for the zero-use contribution.

Lemma 8.6 (Single-bridge stationarity). *One has*

$$P \left(\sum_{i \in I} \frac{\xi_i}{r_i} - 1 \right) + Q \left(\sum_{j \in J} \frac{\zeta_j}{c_j} - 1 \right) = \frac{\tau H + \sum_{k \geq 1} k H_k}{1 - \tau}. \quad (8.27)$$

Moreover,

$$H_0 \geq L_{7,a,b}^\sharp (1-t)^7 e^{s/2}. \quad (8.28)$$

Proof. Multiply the entries of $A[I, J]$ by a parameter and globally renormalize to keep the total entry sum equal to seven. This is a two-sided feasible curve. Every permutation using k entries of $I^c \times J^c$ uses exactly $k+1$ core entries. Differentiating Φ at the maximizing matrix gives (8.27).

For the second assertion, Theorem 3.12 and (8.25) give

$$H_0 \geq q^7 \text{per}(B) \geq L_{7,a,b}^\sharp q^7 e^{s/2}.$$

Finally $q = 1 - e - f + x \geq 1 - t$. □

For brevity write

$$L = L_{7,a,b}^\sharp, \quad A_0 = L(1-t)^7, \quad (8.29)$$

and define

$$\mathcal{B} = P_a(e) \frac{e}{m-e} + P_b(f) \frac{f}{\ell-f} - \frac{(t+d)K - dA_0}{1-t}. \quad (8.30)$$

Lemma 8.7 (Stationarity–entropy squeeze). *At a feasible single-bridge maximizer satisfying (8.18), one has*

$$H \geq A_0 e^{s/2}, \quad (8.31)$$

and, whenever $\mathcal{B} > 0$,

$$su \geq \frac{\mathcal{B}^2}{3}. \quad (8.32)$$

On the other hand,

$$su \leq \frac{(K - A_0)^2}{2A_0}. \quad (8.33)$$

Proof. The first assertion is (8.28) and $H \geq H_0$. By the harmonic-mean inequality,

$$\sum_{i \in I} \frac{\xi_i}{r_i} - 1 \geq \frac{e}{m-e} - \sqrt{s_I V_I},$$

and similarly in the column group. Since a permutation uses at most d entries of $I^c \times J^c$,

$$\sum_{k \geq 1} k H_k \leq d(H - H_0).$$

The right side of (8.27) is therefore at most

$$\frac{(\tau + d)(K - u) - dA_0}{1 - \tau} \leq \frac{(t + d)(K - u) - dA_0}{1 - t}. \quad (8.34)$$

Indeed, the derivative of the first rational function with respect to τ is

$$\frac{(1 + d)(K - u) - dA_0}{(1 - \tau)^2} > 0,$$

because $K - u = H \geq H_0 \geq A_0$, and then $\tau \leq t$.

Put

$$E = \frac{e}{m - e}, \quad F = \frac{f}{\ell - f}, \quad c = \frac{t + d}{1 - t},$$

and denote the common value in (8.27) by Z . The harmonic-mean estimates and $P, Q \leq 1$ give

$$PE + QF - Z \leq \sqrt{s_I V_I} + \sqrt{s_J V_J}. \quad (8.35)$$

On the other hand, (8.34) and $P = P_a - u_r$, $Q = P_b - u_c$ give

$$PE + QF - Z \geq \mathcal{B} + cu - u_r E - u_c F.$$

Here $c \geq \max\{E, F\}$, because $E \leq e/(1 - e) \leq t/(1 - t) < c$, and similarly for F . Together with $u \geq u_r + u_c$, this proves $PE + QF - Z \geq \mathcal{B}$. Lemma 8.5 and Cauchy–Schwarz now yield

$$\mathcal{B} \leq \sqrt{s_I V_I} + \sqrt{s_J V_J} \leq \sqrt{3s(u_r + u_c)} \leq \sqrt{3su}.$$

This proves (8.32).

Finally, $K - u = H \geq A_0 e^{s/2}$ and $e^{s/2} \geq 1 + s/2$ imply

$$su \leq \frac{2u(K - A_0 - u)}{A_0} \leq \frac{(K - A_0)^2}{2A_0},$$

which is (8.33). □

8.5 The exact two-variable certificate

All quantities in the preceding subsection are rational functions of e, f . On the square

$$\mathcal{Q} = \left[0, \frac{3}{20}\right]^2$$

the denominator

$$\Delta = (m - e)(\ell - f)(1 - e - f)$$

is positive. Define the three rational-coefficient polynomials

$$\mathcal{G} = A_0 - K, \quad \mathcal{N} = \Delta \mathcal{B},$$

and

$$\mathcal{F} = 2A_0 \mathcal{N}^2 - 3(K - A_0)^2 \Delta^2. \quad (8.36)$$

If $\mathcal{G} > 0$, then (8.31) contradicts $H = K - u \leq K$. If $\mathcal{N} > 0$ and $\mathcal{F} > 0$, then (8.32) and (8.33) contradict each other. Thus it remains only to certify the following finite polynomial alternative.

Proposition 8.8 (Exact single-bridge certificate). *For each $(a, b) \in \{(1, 5), (2, 4), (3, 3)\}$ and every $(e, f) \in \mathcal{Q}$,*

$$\mathcal{G}(e, f) > 0 \quad \text{or} \quad (\mathcal{N}(e, f) > 0 \text{ and } \mathcal{F}(e, f) > 0). \quad (8.37)$$

Proof. The proof is an exact tensor-product Bernstein certificate. On a rectangle, a polynomial is written in the Bernstein basis of its bidegree. The basis functions are nonnegative and sum to one, so positivity of every coefficient certifies strict positivity on that rectangle. Starting from $\mathcal{Q}$, the supplementary script subdivides the physically longer side at its midpoint whenever neither branch in (8.37) is yet coefficientwise positive. All coefficients and all subdivision endpoints are instances of `Fractions.Fraction`; no floating-point operation or tolerance is used.

The resulting finite trees are

(a, b)	nodes	$\mathcal{G}$ -leaves	$(\mathcal{N}, \mathcal{F})$ -leaves	maximum depth
$(1, 5)$	39	10	10	8
$(2, 4)$	35	9	9	6
$(3, 3)$	47	11	13	8.

(8.38)

The complete coefficient generation and de Casteljau subdivision are given in the supplementary certificate described in Appendix B. $\square$

Theorem 8.9 (Boundary exclusion in dimension seven). *No global maximizer of Φ on K_7 has a zero entry.*

Proof. The case $\tau = 0$ is excluded after Lemma 5.5. For $\tau > 0$, Proposition 8.3 excludes one-sided pairs, Proposition 8.1 excludes two-sided pairs with $p \geq 2$, and Proposition 8.4 excludes single-bridge pairs with a nonpositive complementary excess. In the remaining case, Lemma 8.7 and Proposition 8.8 give a contradiction. Hence no boundary global maximizer exists. $\square$

9 Boundary exclusion in dimension six

We now treat the remaining dimension. Throughout this section,

$$n = 6, \quad \gamma = \gamma_6 = \frac{5}{324}, \quad \kappa = \kappa_6 = v_5 G(5) = \frac{6144}{390625} > \gamma. \quad (9.1)$$

Let A be a hypothetical boundary global maximizer, and retain the notation

$$R = \prod_i r_i = 1 - \rho, \quad C = \prod_j c_j = 1 - \sigma, \quad H = \text{per}(A) = \gamma - \delta$$

from Section 5.

9.1 Uniform localization

Lemma 9.1 (Dimension-six localization). *For every nonempty proper set E of rows, if*

$$\sum_{i \in E} r_i = |E| + e,$$

then

$$|e|^2 \leq \frac{15}{319} < \left(\frac{11}{50}\right)^2 < \left(\frac{7}{32}\right)^2. \quad (9.2)$$

The same assertion holds for every column subset. In particular,

$$r_i, c_j > \frac{25}{32} \quad (1 \leq i, j \leq 6). \quad (9.3)$$

Proof. If $e > 0$, separate AM–GM on E and E^c , followed by binary Pinsker, gives

$$\frac{2e^2}{6} \leq -\log(1 - \rho) \leq \frac{\rho}{1 - \rho} \leq \frac{\gamma}{1 - \gamma}.$$

Thus $e^2 \leq 15/319$. If $e < 0$, the complementary row set has positive excess $-e$. The column proof is identical. The rational comparisons in (9.2) are verified exactly in the supplementary certificate. $\square$

Set

$$E_0 = \frac{11}{50}. \quad (9.4)$$

As in the preceding dimensions, a boundary maximizer with Hall deficit $\tau = 0$ is already doubly stochastic and has a zero. Corollary 3.6 would then give $H \geq \kappa > \gamma$, contradicting Lemma 5.1. Hence $\tau > 0$.

9.2 Two-sided critical Hall pairs

Assume first that a critical pair has $a, b \geq 1$, and transpose so that $a \leq b$. The six possible types are

$$(a, b, p) = (1, 1, 4), (1, 2, 3), (1, 3, 2), (2, 2, 2), (1, 4, 1), (2, 3, 1), \quad (9.5)$$

where $p = 6 - a - b$. Put

$$e = \sum_{i \in I^c} r_i - a, \quad f = \sum_{j \in J^c} c_j - b, \quad x = s_A(I^c, J^c), \quad t = e + f. \quad (9.6)$$

Criticality gives

$$p\tau = e + f - x = t - x. \quad (9.7)$$

Consequently,

$$-E_0 \leq e, f \leq E_0, \quad 0 < t \leq 2E_0 = \frac{11}{25} < p, \quad \tau \leq \frac{t}{p}. \quad (9.8)$$

The critical block is saturated:

$$B[I, J] = \frac{A[I, J]}{1 - \tau}.$$

Let $m = 6 - a$, $\ell = 6 - b$, and let ξ, η be the normalized row and column marginals of this core, as in Proposition 3.14. Put

$$S_\xi = \sum_{i \in I} \left(\xi_i - \frac{1}{m} \right)^2, \quad S_\eta = \sum_{j \in J} \left(\eta_j - \frac{1}{\ell} \right)^2. \quad (9.9)$$

Decompose the permanent as

$$H = H_0 + H_1 + \cdots + H_d, \quad d = \min(a, b) = a,$$

where H_k is the contribution of permutations using exactly k entries of $I^c \times J^c$. Such a permutation uses exactly $p + k$ core entries.

Lemma 9.2 (Dimension-six core stationarity). *At the global maximizer,*

$$R \left(\sum_{i \in I} \frac{\xi_i}{r_i} - 1 \right) + C \left(\sum_{j \in J} \frac{\eta_j}{c_j} - 1 \right) = \frac{\tau H + p^{-1} \sum_{k \geq 1} k H_k}{1 - \tau}. \quad (9.10)$$

Consequently,

$$R \left(\sum_{i \in I} \frac{\xi_i}{r_i} - 1 \right) + C \left(\sum_{j \in J} \frac{\eta_j}{c_j} - 1 \right) \leq \frac{(t + d)H - dH_0}{p - t}. \quad (9.11)$$

Proof. Multiply the core $A[I, J]$ by a parameter and globally renormalize so that the total entry sum remains six. Its mass is $p(1 - \tau)$. Dividing the first variation by this mass, the row-product contribution is

$$R \left(\sum_{i \in I} \frac{\xi_i}{r_i} - 1 \right),$$

and the column contribution is analogous. A monomial in H_k uses $p + k$ core entries, so after global normalization its divided derivative is $(p\tau + k)/(p(1 - \tau))$. Stationarity proves (9.10).

Since $k \leq d$,

$$\sum_{k \geq 1} k H_k \leq d(H - H_0).$$

Writing $z = p\tau$, the resulting upper bound is $((z + d)H - dH_0)/(p - z)$, an increasing function of z because $(p + d)H - dH_0 > 0$. Use $z \leq t$. $\square$

Let $L = L_{6,a,b}^\sharp$. Define c_ξ, c_η by

$$(c_\xi, c_\eta) = \begin{cases} \left(\frac{p^2}{2(a^2 + b^2)}, \frac{p^2}{2(a^2 + b^2)} \right), & p \geq 2, \\ \left(\frac{1}{2} + \frac{1}{3m}, \frac{1}{2} + \frac{1}{3\ell} \right), & p = 1. \end{cases} \quad (9.12)$$

Put

$$A_0 = L \left(1 - \frac{t}{p} \right)^6. \quad (9.13)$$

Because $A \geq (1 - \tau)B$, Proposition 3.14 and (9.8) give

$$H_0 \geq A_0(1 + c_\xi S_\xi + c_\eta S_\eta). \quad (9.14)$$

The joint deficit budget also gives

$$H \leq \overline{H} := R + C - 2 + \gamma. \quad (9.15)$$

It follows from Lemma 9.2 that both

$$F = \frac{(t+d)\overline{H} - dA_0(1 + c_\xi S_\xi + c_\eta S_\eta)}{p-t} - R \left(\sum_{i \in I} \frac{\xi_i}{r_i} - 1 \right) - C \left(\sum_{j \in J} \frac{\eta_j}{c_j} - 1 \right), \quad (9.16)$$

$$G = \overline{H} - A_0(1 + c_\xi S_\xi + c_\eta S_\eta) \quad (9.17)$$

are nonnegative. Indeed, in (9.11) replace H by the larger quantity $\overline{H}$ and H_0 by the lower bound (9.14); this gives $F \geq 0$. Moreover $\overline{H} \geq H \geq H_0$, which gives $G \geq 0$. Hence $F + \zeta G \geq 0$ for every $\zeta \geq 0$.

9.3 A two-level extremal reduction

For a group of q nonnegative numbers z_i with total s , a probability vector w , and constants $K, B_*, \lambda > 0$, define polynomially on the closed product of simplices

$$\mathcal{T}(z, w) = K \prod_{i=1}^q z_i \left(B_* - \sum_{i=1}^q \frac{w_i}{z_i} \right) - \lambda \sum_{i=1}^q \left(w_i - \frac{1}{q} \right)^2. \quad (9.18)$$

Lemma 9.3 (Two-level side maximizer). *Suppose $s/q > 25/32$ and $B_* > 32/25$. The maximum of (9.18) is attained in the following form. For some $k \in \{1, \dots, q\}$,*

$$w_i = \frac{1}{k} \quad (1 \leq i \leq k), \quad w_i = 0 \quad (i > k),$$

and the z_i have two levels,

$$z_i = h \quad (i \leq k), \quad z_i = \ell_0 \quad (i > k), \quad h \geq \ell_0.$$

For $k < q$ they can be parametrized by $0 \leq u \leq 1$ as

$$h = \frac{s}{q} \left(1 + u \frac{q-k}{k} \right), \quad \ell_0 = \frac{s}{q} (1 - u). \quad (9.19)$$

For $k = q$, one has $h = s/q$.

Proof. The maximum exists. If one of the z_i vanishes, the polynomial form of (9.18) is nonpositive. At the uniform point $z_i = s/q$, $w_i = 1/q$, its value is $K(s/q)^q (B_* - q/s) > 0$. Thus every maximizing z_i is positive, the maximum is positive, and at a maximizer $B_* - \sum_i w_i/z_i > 0$.

Choose among all maximizers one minimizing

$$\mathcal{V} = \sum_i z_i^2 + \sum_i w_i^2.$$

For two indices preserve their two z - and w -sums and write

$$z_i = h + x, \quad z_j = h - x, \quad w_i = q_0 + y, \quad w_j = q_0 - y.$$

Up to an additive constant, the pair-dependent part is

$$-P_0Ax^2 + 2P_0xy - 2\lambda y^2, \quad (9.20)$$

where $P_0 > 0$ and

$$A = B_* - \sum_{r \neq i, j} \frac{w_r}{z_r} > 0.$$

If both w_i, w_j are positive, the pair variables are interior. The stationarity equations imply either $x = y = 0$, or $y = Ax$ and $P_0 = 2\lambda A$. In the latter case (9.20) equals $-(P_0/A)(Ax - y)^2$, so the segment from (x, y) to $(0, 0)$ is flat. Averaging the pair would preserve the maximum and strictly decrease $\mathcal{V}$, a contradiction. Hence all positive coordinates of w are equal, and their corresponding z_i are equal.

If both w_i, w_j vanish, only x varies and (9.20) is $-P_0Ax^2$; hence all inactive z_i are equal. Since w is a probability vector, its k positive coordinates are $1/k$. Finally, moving an infinitesimal amount of w from an active coordinate to an inactive one gives

$$K \prod_i z_i \left(\frac{1}{h} - \frac{1}{\ell_0} \right) + \frac{2\lambda}{k} \leq 0,$$

so $h > \ell_0$ when $k < q$. The parametrization follows from the fixed total. $\square$

Set

$$\zeta = \begin{cases} 14, & p \geq 2, \\ 8, & p = 1, \end{cases} \quad (9.21)$$

and define

$$B_* = \frac{p+d}{p-t} + \zeta, \quad \lambda_\xi = c_\xi A_0 \left(\frac{d}{p-t} + \zeta \right), \quad \lambda_\eta = c_\eta A_0 \left(\frac{d}{p-t} + \zeta \right). \quad (9.22)$$

By (9.3), $\sum_i \xi_i/r_i < 50/39 < B_*$, and similarly for columns. AM–GM on the complementary groups therefore increases the upper bound for $F + \zeta G$ when R, C are replaced by

$$\hat{R} = \left(1 + \frac{e}{a}\right)^a \prod_{i \in I} r_i, \quad \hat{C} = \left(1 + \frac{f}{b}\right)^b \prod_{j \in J} c_j.$$

Consequently,

$$F + \zeta G \leq \mathcal{U} = C_0(t) + T_a(e, r, \xi) + T_b(f, c, \eta), \quad (9.23)$$

where

$$C_0(t) = \frac{(t+d)(\gamma-2) - dA_0}{p-t} + \zeta(\gamma-2-A_0), \quad (9.24)$$

$$T_a = \left(1 + \frac{e}{a}\right)^a \prod_{i \in I} r_i \left(B_* - \sum_{i \in I} \frac{\xi_i}{r_i} \right) - \lambda_\xi S_\xi, \quad (9.25)$$

and T_b is the analogous column expression.

For a side of complement size a , group size $m = 6 - a$, support size k , and excess e , Lemma 9.3 gives

$$T_a \leq \left(1 + \frac{e}{a}\right)^a h^{k-1} \ell_0^{m-k} (B_* h - 1) - \lambda_\xi \left(\frac{1}{k} - \frac{1}{m} \right), \quad (9.26)$$

where h, ℓ_0 are given by (9.19) with $s = m - e$. The formula for $k = m$ omits the low level and entropy term. The column side is identical.

9.4 The exact four-variable certificate

Map the triangular domain (9.8) to the unit square by

$$t = 2E_0x, \quad e = t - E_0 + z(2E_0 - t), \quad f = E_0 - z(2E_0 - t), \quad 0 \leq x, z \leq 1. \quad (9.27)$$

For every type (a, b) , every pair of support sizes

$$1 \leq k \leq 6 - a, \quad 1 \leq k' \leq 6 - b,$$

and the two spread parameters $u, v \in [0, 1]$, substitute (9.19), (9.27), and (9.21) into the right side of (9.23). Define

$$P_{a,b,k,k'}(x, z, u, v) = -(p - t)\mathcal{U}. \quad (9.28)$$

This is a rational polynomial. Exact tensor-product Bernstein subdivision proves

$$P_{a,b,k,k'} > 0 \quad \text{on } [0, 1]^4 \quad (9.29)$$

for all 98 cases. The certificate summary is

(a, b)	polynomials	Bernstein nodes	maximum depth
(1, 1)	25	165	6
(1, 2)	20	276	7
(1, 3)	15	483	14
(2, 2)	16	416	13
(1, 4)	10	352	14
(2, 3)	12	558	15

(9.30)

Every terminal Bernstein coefficient is a strictly positive rational number. The smallest terminal coefficient for the tightest type (2, 3) is

$$\frac{206812176444269171}{2388787200000000000000} > 0. \quad (9.31)$$

The supplementary script constructs every polynomial from (9.28), performs exact power-to-Bernstein conversion, and uses exact de Casteljau subdivision at $1/2$.

Since $p - t > 0$, (9.29) gives $\mathcal{U} < 0$, contradicting

$$0 \leq F + \zeta G \leq \mathcal{U}.$$

Thus no two-sided critical Hall pair can occur.

9.5 One-sided critical Hall pairs

After transposition, assume $I = [6]$ and $|J| = p = 6 - b$. Then

$$\sum_{j \in J} c_j = p(1 - \tau), \quad 0 < \tau < \frac{E_0}{p}. \quad (9.32)$$

The corresponding columns of B saturate $A/(1 - \tau)$. Put

$$m_i = \sum_{j \in J} b_{ij}, \quad \beta_i = \frac{m_i}{p}, \quad s = \sum_i \left(\beta_i - \frac{1}{6} \right)^2.$$

Lemma 7.3 gives

$$D := 1 - \sum_i \frac{\beta_i}{r_i} \geq \frac{(1-\gamma)\tau}{1-\tau} > 0. \quad (9.33)$$

Lemma 9.4 (Dimension-six row energy). *With $y_i = 1 - r_i^{-1}$,*

$$\rho > \frac{3}{10} \sum_i y_i^2. \quad (9.34)$$

Consequently,

$$\rho > \frac{3}{10} \frac{(1-\gamma)^2 \tau^2}{(1-\tau)^2 s}. \quad (9.35)$$

Proof. By Lemma 9.1, $r_i > 25/32$. The integral argument in Lemma 7.4 gives

$$r - 1 - \log r \geq \frac{(25/32)^2}{2} \left(1 - \frac{1}{r}\right)^2.$$

Since $1 - \rho \geq 1 - \gamma$,

$$\rho \geq (1 - \rho)(-\log(1 - \rho)) \geq \frac{(1 - \gamma)(25/32)^2}{2} \sum_i y_i^2.$$

The exact gap

$$\frac{(1 - \gamma)(25/32)^2}{2} - \frac{3}{10} = \frac{1547}{3317760} > 0 \quad (9.36)$$

proves (9.34). The harmonic-mean inequality gives $\sum_i y_i \leq 0$, and Cauchy–Schwarz together with (9.33) proves (9.35). In particular, $s > 0$. $\square$

Retain the exact complementary column-product loss. Two-group AM–GM gives

$$\sigma \geq 1 - Q_b(\tau), \quad Q_b(\tau) = \left(1 + \frac{p\tau}{b}\right)^b (1 - \tau)^p. \quad (9.37)$$

Because A is a boundary matrix, the dominated doubly stochastic matrix B has a zero. If $q^{(j)}$ are the p saturated columns, then convexity of squared distance gives

$$s \leq \frac{1}{p} \sum_{j \in J} s_6(q^{(j)}) \leq \max_{j \in J} s_6(q^{(j)}).$$

Apply Theorem 3.9 to the maximizing column and Corollary 3.6. Since $A \geq (1 - \tau)B$, we obtain

$$H \geq (1 - \tau)^6 \max\{\kappa, \gamma \exp(s/2)\}. \quad (9.38)$$

On the other hand,

$$H \leq \gamma - \rho - \sigma. \quad (9.39)$$

Set $s_0 = 2 \log(\kappa/\gamma)$.

If $s \leq s_0$, then $s_0 \leq 2(\kappa - \gamma)/\gamma$, and (9.35) gives

$$\rho > A \frac{\tau^2}{(1 - \tau)^2}, \quad A = \frac{3}{10} \frac{(1 - \gamma)^2 \gamma}{2(\kappa - \gamma)} = \frac{39750390625}{5253139008}. \quad (9.40)$$

It is therefore enough to prove

$$(1 - \tau)^2 [1 - Q_b(\tau) + \kappa(1 - \tau)^6 - \gamma] + A\tau^2 > 0. \quad (9.41)$$

For $b = 1, \dots, 5$, exact univariate Bernstein subdivision proves this on $0 \leq \tau \leq 11/(50p)$; the required depths are respectively 1, 2, 3, 4, 5.

If $s \geq s_0$, use $\exp(s/2) \geq 1 + s/2$ and AM–GM in (9.35). With $q = 1 - \tau$,

$$\rho + \gamma q^6 \exp(s/2) - \gamma > (1 - \gamma)\tau q^2 \sqrt{\frac{3\gamma}{5}} + \gamma(q^6 - 1).$$

Here $3\gamma/5 = 1/108$, and

$$\frac{(1 - \gamma)^2}{108} - \frac{9}{1024} = \frac{33853}{181398528} > 0. \quad (9.42)$$

Thus it remains to prove, for $\tau > 0$,

$$1 - Q_b(\tau) + \frac{3}{32}\tau(1 - \tau)^2 + \gamma((1 - \tau)^6 - 1) > 0. \quad (9.43)$$

The left side has a factor τ . For all five values of b , the quotient has globally positive Bernstein coefficients on $[0, 11/(50p)]$; the smallest coefficient is $1/864$.

Both entropy regimes contradict (9.38)–(9.39). Hence no one-sided critical pair exists.

Theorem 9.5 (Boundary exclusion in dimension six). *No global maximizer of Φ on K_6 has a zero entry.*

Proof. The case $\tau = 0$ was excluded after Lemma 9.1. If $\tau > 0$, a critical pair exists. The preceding subsections exclude both its two-sided and one-sided forms. Thus no boundary global maximizer exists. $\square$

10 Proof of the main theorem and corollaries

Proof of Theorem 2.1. Fix $6 \leq n \leq 15$, and let A be a global maximizer of Φ on K_n . Theorem 9.5 applies for $n = 6$, Theorem 8.9 applies for $n = 7$, Theorem 7.7 applies for $8 \leq n \leq 10$, and Theorem 6.4 applies for $11 \leq n \leq 15$. Hence A is positive. Theorem 4.5 gives $A = U_n$.

At U_n , all row sums and column sums are one, and

$$\text{per}(U_n) = \frac{n!}{n^n}.$$

Therefore

$$\max_{A \in K_n} \Phi(A) = \Phi(U_n) = 2 - \frac{n!}{n^n}.$$

Any equality case is itself a global maximizer and is therefore U_n . $\square$

Corollary 10.1 (No boundary maximizers). *For every $6 \leq n \leq 15$, every global maximizer of Φ lies in the relative interior of K_n .*

Corollary 10.2 (Sharp zero-block permanent inequality). *If a doubly stochastic $n \times n$ matrix has an $a \times b$ zero block with $a + b \leq n - 1$, then*

$$\text{per}(B) \geq \frac{v_{n-a}v_{n-b}}{v_{n-a-b}},$$

and equality is attainable by a block-constant doubly stochastic matrix.

11 Discussion and the remaining case

The proof has several logically separate modules. Stable-polynomial capacity first yields the support-sensitive permanent inequality, the sharp rectangular-zero-block bound, and three entropy refinements. The exact complement entropy Ψ_m controls both a selected supported column and the weighted geometric mean of all complementary minors. These two forms combine into the core-marginal bound of Proposition 3.14.

The positive-maximizer theorem is independent of the boundary analysis. Pairwise averaging is flat at a positive global maximizer, while the Hessian at the uniform matrix is negative definite on the tangent hyperplane. Cyclic averaging therefore identifies the unique interior maximizer.

The common boundary framework consists of the joint deficit budget, relative-entropy subset estimates, Hall dilation, and support-sensitive permanent lower bounds. Scalar first- and second-order comparisons settle dimensions eight through fifteen and all ordinary dimension-seven cuts. The three dimension-seven single-bridge types require complementary-minor entropy, a core stationarity identity, and conditional product energy; the last separation is the exact two-variable alternative in Proposition 8.8.

In dimension six, marginal entropy must be retained even when the Hall core has size $p \geq 2$. The exact stationarity identity in Lemma 9.2 and the two-level reduction in Lemma 9.3 turn every two-sided configuration into one of 98 four-variable polynomial inequalities. The one-sided argument is sharper than its higher-dimensional counterpart because it keeps the full complementary column-product envelope $Q_b(\tau)$.

The computer-assisted component consists of finite exact rational certificates: scalar comparisons in all dimensions, the dimension-seven one- and two-variable Bernstein certificates, the 98 dimension-six four-variable certificates, and ten dimension-six univariate certificates. The analytic reductions, the polynomials being certified, the positivity criterion, and all subdivision statistics are stated in the text. No floating-point comparison, tolerance, random choice, or numerical optimizer is used.

Combined with the results cited in the introduction, the present theorem leaves only dimension 5. The dimension-six argument suggests that a future treatment of that case should retain still more of the transportation geometry of a critical Hall core, rather than seek a purely scalar permanent lower bound.

Verification status. No analytic step in the proof is intentionally deferred to an external theorem of Gurvits or Hwang. Standard results used explicitly are AM–GM, max-flow/min-cut, Hurwitz’s theorem, Gauss–Lucas, and elementary Taylor expansion. The finite comparisons are exact rational computations whose formulas are recorded below and whose complete source code is supplied as supplementary material. They should nevertheless be independently checked by a human referee or formalized in a proof assistant before publication; the present manuscript is not a machine-checked formal proof.

A Exact rational certificates

This appendix records representative exact comparisons used in the text. The supplementary programs verify every scalar margin and every Bernstein subdivision.

For dimension six,

$$\gamma_6 = \frac{5}{324}, \quad \kappa_6 = \frac{6144}{390625}, \quad \frac{15}{319} < \left(\frac{11}{50}\right)^2 < \left(\frac{7}{32}\right)^2.$$

The row-energy and large-entropy gaps are

$$\frac{(1 - \gamma_6)(25/32)^2}{2} - \frac{3}{10} = \frac{1547}{3317760}, \quad \frac{(1 - \gamma_6)^2}{108} - \frac{9}{1024} = \frac{33853}{181398528}.$$

The four-variable certificate is summarized in (9.30), and the smallest recorded terminal coefficient for its tightest type is (9.31). The one-sided certificates consist of five small-entropy polynomials and five large-entropy quotients; the latter have minimum Bernstein coefficient $1/864$.

For dimension seven, the localization and half-gap comparisons are

$$\eta_7^2 - \frac{\gamma_7}{14(1 - \gamma_7)} = \frac{23263}{1808073127}, \quad \frac{1}{2} - \frac{3}{7} - \eta_7 = \frac{33}{658}. \quad (\text{A.1})$$

The minimum ordinary two-sided margin is

$$\frac{217499955444827479495}{21713873952987016007066496},$$

and the minimum sign-mixed single-bridge margin is

$$\frac{165035926385}{3897251304526692}.$$

The one-sided energy, small-entropy, and large-entropy certificates are the positive quantities in (8.8), (8.10), and (8.13). The exact two-variable certificate is summarized in (8.38).

The remaining exact margins are displayed in equations (6.3), (6.13), (6.14), (7.5), (7.23), and (7.26).

For the localization constants in (5.9), one has

n	$\eta_n^2 - \frac{\gamma_n}{2n(1 - \gamma_n)}$	$\frac{1}{2} - \frac{\lfloor (n-1)/2 \rfloor}{n} - \eta_n$
7	$\frac{23263}{1808073127}$	$\frac{33}{658}$
8	$\frac{4757}{836844800}$	$\frac{9}{80}$
9	$\frac{242489}{87087962025}$	$\frac{13}{270}$
10	$\frac{31109}{41313127850}$	$\frac{11}{115}$
11	$\frac{63007753811}{34945789841847500}$	$\frac{82}{1925}$
12	$\frac{11651173}{90828753405000}$	$\frac{319}{3900}$
13	$\frac{13072911571453}{366471344281458130000}$	$\frac{537}{14300}$
14	$\frac{71438752181}{2511343447066440000}$	$\frac{893}{12600}$

(A.2)

Every entry is positive.

The auxiliary comparison used for $11 \leq n \leq 14$ is

$$\frac{1}{1000} - \gamma_{11} = \frac{22308624601}{25937424601000} > 0. \quad (\text{A.3})$$

For $n = 8, 9, 10$, the inequalities $\gamma_n < 1/81$ follow from

$$\begin{array}{c|c} n & 1/81 - \gamma_n \\ \hline 8 & \frac{105557}{10616832} \\ 9 & \frac{54569}{4782969} \\ 10 & \frac{1516573}{126562500}. \end{array} \quad (\text{A.4})$$

For dimension 15, the six inequalities in (6.19)–(6.21) are equivalent to the positivity of the following integers:

$$\begin{aligned} 15^{15} - 300000 \cdot 15! &= 45\,591\,579\,980\,859\,375, \\ 29863 \cdot 15^{15} - 10^{10} 15! &= 81\,568\,443\,603\,515\,625, \\ 10^{10} 13! 13^{13} - 29937 \cdot 14^{26} &= 290\,584\,966\,207\,055\,614\,607\,247\,802\,368, \\ 3 \cdot 14^{26} - 10^6 13! 13^{13} &= 3\,939\,835\,293\,455\,120\,947\,239\,518\,208, \\ 10^8 13! 12^{24} - 301 \cdot 13^{37} &= 18\,198\,910\,159\,813\,162\,803\,331\,251\,792\,701\,065\,161\,767, \\ 4 \cdot 13^{37} - 10^6 13! 12^{24} &= 162\,574\,843\,672\,041\,436\,633\,515\,011\,954\,046\,933\,353\,332. \end{aligned} \quad (\text{A.5})$$

B Exact-arithmetic verification

The supplementary package contains three executable files:

<code>dittert_n6_certificates.py</code>	dimension-six polynomial generation and certificates,
<code>dittert_n7_n15_certificates.py</code>	dimensions six through fifteen,
<code>run_all_certificates.py</code>	a wrapper that runs both verifiers.

Run

```
python3 run_all_certificates.py.
```

A successful run ends with

```
All Dittert certificates for dimensions 6 through 15 passed.
```

The dimension-seven-through-fifteen verifier requires only Python 3 and the standard library. It evaluates every localization, Hall-range, first-order, and second-order margin as a `Fraction`; constructs the univariate and bivariate dimension-seven polynomials; performs exact power-to-Bernstein conversion and exact dyadic de Casteljau subdivision; and checks all reported tree statistics literally.

The dimension-six verifier was tested with Python 3.13, SymPy 1.14, and NumPy 2.3. SymPy is used only to expand the explicit rational polynomials, and NumPy only stores object arrays. Every generated coefficient is converted to `fractions.Fraction` before Bernstein conversion,

subdivision, or comparison. The program constructs all 98 polynomials in (9.28), checks the node and depth counts in (9.30), and verifies the ten one-sided polynomials. The accompanying `requirements.txt` records the tested dependencies.

Neither verifier contains a floating-point comparison, tolerance, random choice, or numerical optimization routine. The Bernstein implication is elementary: on a box, tensor-product Bernstein basis functions are nonnegative and sum to one, so a polynomial lies between the smallest and largest of its Bernstein coefficients.

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